European Journal of Medicinal Chemistry 242 (2022) 114681

Contents lists available at ScienceDirect

European Journal of Medicinal Chemistry

journal homepage: www.elsevier.com/locate/ejmech

Research paper

(1*R*,3*S*)-THCCA-Asn: To show the discovery of selective inhibitor of thrombin by successfully combining virtual screening and biological assay

Xiaoyi Zhang [1], Yifan Yang [1], Yuji Wang, Jianhui Wu, Qiongyan Fang, Yaonan Wang, Ming Zhao [**], Shiqi Peng [*]

College of Pharmaceutical Sciences, Capital Medical University, Beijing, PR China

ARTICLE INFO

Keywords:
Virtual screening
Thrombin inhibitor
Anti-platelet aggregation
Anti-thrombotic activity

ABSTRACT

Thrombin is the most potent platelet aggregator. To discover the selective inhibitor of thrombin that is important to curing platelet aggregation-related diseases, docking experiments were performed to dock (1*R*,3*S*)-2,3,4,9-tetrahydro-β-carboline-3- carboxylic acid, [(1*R*,3*S*)-THCCA], and (1*S*,3*S*)-2,3,4,9-tetrahydro-β-carboline-3- carboxylic acid, [(1*S*,3*S*)-THCCA], into the p pocket of bovine thrombin. The ideal match supported that (1*R*,3*S*)-THCCA could be used as a potential lead compound. In this case 20 natural amino acids were theoretically introduced into the 3-carboxyl of (1*R*,3*S*)-THCCA and 20 derivatives, (1*R*,3*S*)-THCCA-amino acids, were docked into p pocket of bovine thrombin to perform virtual screening. The screening revealed that comparing to (1*R*,3*S*)-THCCA itself the DockScores of 16 derivatives were higher, and (1*R*,3*S*)-THCCA-Asn (**4j**) got the highest DockScore. Thus, 16 derivatives were synthesized for experimental study. The *in vitro* anti-platelet aggregation assay showed that at 100 μM of concentration the 16 derivatives failed to inhibit the platelet aggregation induced by both adenosine diphosphate and arachidonic acid. On the other hand, however, the IC_{50} value of the 16 derivatives inhibiting the platelet aggregation induced by platelet activating factor and thrombin ranged from 9.44 μM to 194.64 μM and from 0.07 μM to 9.56 μM, respectively. The *in vitro* anti-platelet aggregation assay suggested that the 16 derivatives selectively inhibited the platelet aggregation induced by thrombin. In particular, the IC_{50} of (1*R*,3*S*)-THCCA-Asn (**4j**) had the lowest value. On rat model at 1 nmol/kg of dosage the 16 derivatives effectively prevented thrombus formation. It is worth pointing out that even at 0.01 nmol/kg of dosage, **4j** still effectively prevented thrombus formation. **4j** hardly has effects on the proliferation of mammalian cells and rat tail bleeding time. In conclusion, the combination of virtual screening and biological assays successfully lead to the discovery of **4j** as a promising candidate of selective inhibitor of thrombin.

1. Introduction

So far thrombin is known as the last one of trypsin-like plasma serine proteases family and is the most potent one of all platelet aggregators [1,2]. Due to its ability of catalyzing the conversion of fibrinogen, activating FXIII and inducing the platelet aggregation, thrombin is the key enzyme for the cascade of blood coagulation [3,4]. In these cases, the inhibition of thrombin generation and activity will benefit patients with acute coronary syndromes [5–7]. The discovery of highly selective and orally available thrombin inhibitors has attracted efforts in recent years. In this context, some heterocycles are considered as the promising leads such as tetrahydroiso-quinoline [8], 3,4-dihydro-2*H*-1,4-benzoxazi- ne [9], imidazo[1,5-*c*]imidazole-3-ones [10], polysulfated xanthones [11], ketopiperazino methylazaindole [12,13], pyrazinone [14–16], 3-amidinobenzyl-1*H*-indole-2-carboxami- des [17,18], benzothiophenes [19], bis(phenyl)methane [20,21] and flavonoids [22–24].

In the discovery of thrombin inhibitors the X-ray based crystal structures of the complex of bovine thrombin with benzoxazoles, chlorothiophenyl containing molecules and carboline derivatives are widely used as templates [25–27]. The docking experiments that target bovine thrombin lead to the discovery of a series of thrombin inhibitors [28–33].

The development of antithrombotic derivatives of β-carboline-3-carboxlic acid was the focus of our research [34–50]. To find the

* Corresponding author.
** Corresponding author.
E-mail addresses: maozhao@126.com (M. Zhao), sqpeng@bjmu.edu.cn (S. Peng).
[1] Co-first author.

https://doi.org/10.1016/j.ejmech.2022.114681
Received 27 May 2022; Received in revised form 21 July 2022; Accepted 11 August 2022
Available online 17 August 2022
0223-5234/© 2022 The Author(s). Published by Elsevier Masson SAS. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

selective inhibitor, we performed the first round docking experiments, in which, (1*R*,3*S*)-tetrahydro-β-carboline-3-carboxylic acid [(1*R*, 3*S*)-THCCA] and (1*S*,3*S*)-tetrahydro-β-carboline-3-carboxylic acid [(1*S*, 3*S*)-THCCA] were docked into the p pocket of bovine thrombin (shown in Fig. 1, PDB ID:1KTS). The DockScores of (1*R*,3*S*)-THCCA and (1*S*, 3*S*)-THCCA were 120.9 and 110.1, respectively (also shown in Fig. 1). This means that as the lead compound of the selective inhibitor of thrombin (1*R*,3*S*)-THCCA is more suitable than (1*S*,3*S*)-THCCA.

In previous studies, natural amino acids were commonly used to modify β-carboline-3-carboxlic acid and to enhance its *in vivo* activity [34,35]. This research hence adopted natural amino acids in the modification of (1*R*,3*S*)-THCCA. To develop selective inhibitors of thrombin, the derivatives of 20 natural amino acids modified (1*R*,3*S*)-THCCA were theoretically constructed, the second round docking experiments were performed. The DockScores are listed in Table 1.

As seen, the DockScores of THCCA-Cys, THCCA-His, THCCA-Met and THCCA- Pro ranged from 87.1 to 107.5, and were lower than 120.9 [the DockScore of (1*R*,3*S*)-THCCA], while the DockScores of the other 16 THCCA-amino acids ranged from 130.5 to 185.7, i.e. higher than 120.9. In this case we prepared the 16 (1*R*,3*S*)-THCCA-amino acids, addressed the *in vitro* IC_{50} values of inhibiting the platelet aggregation induced by four inducer [platelet-activating factor (PAF), adenosine diphosphate (ADP), arachidonic acid (AA) and thrombin (TH)], and evaluated the *in vivo* anti-thrombotic activities on rat model.

2. Results and discussion

2.1. Synthesis of [(1R,3S)-THCCA]-amino acids (4a-p)

(1*R*,3*S*)-THCCA-amino acids (**4a-p**) were synthesized via a six-step reaction procedure of Scheme 1. Based on HPLC analysis (see SI document), the Pictet-Spengler condensation of *L*-Trp-OBzl and methyl 5-formyl-salicylate provided a 2/3 mixture of (1*R*,3*S*)-THCCA benzyl ester and (1*S*,3*S*)-THCCA benzyl ester. The configurations of 1-C of them were assigned based on the NOESY spectra. In the spectra (1*S*,3*S*)-THCCA benzyl ester gave positive NOE signal between the 1-H and 3-H, while (1*R*,3*S*)-THCCA benzyl ester gave negative NOE signal between the 1-H and 3-H. These signals suggest that the configuration of 1-C of (1*S*,3*S*)-THCCA benzyl ester and (1*R*,3*S*)-THCCA benzyl ester is S and R, respectively (NOESY spectra were provided as SI document). In the presence of Pd/C the hydrogenation of (1*R*,3*S*)-THCCA benzyl ester (**1**) provided (1*R*,3*S*)-THCCA (**2**) with 93% of yield. Then **2** was coupled with 16 amino acid benzyl esters to form **3a-p** in 17%–41% of yields. In the presence of Pd/C the hydrogenation of **3a-p** provided (1*R*,3*S*)-THCCA-amino acids (**4a-p**) with 34%–68% yields. These data mean that the present six-step reaction procedure could be used to prepare (1*R*,3*S*)-THCCA-amino acids (**4a-p**) with acceptable yields.

Table 1
DockScores of (1*R*,3*S*)-THCCA-amino acids in p pocket of bovine thrombin.

(1*R*,3*S*)-THCCA-amino acid	DockScore	(1*R*,3*S*)-THCCA-amino acid	DockScore
THCCA-Asn	185.7	THCCA-Glu	140.8
THCCA-Tyr	159.7	THCCA-Phe	135.9
THCCA-Ser	132.5	THCCA-Asp	150.7
THCCA-Lys	156.7	THCCA-Gly	140.6
THCCA-Arg	164.8	THCCA-Thr	130.5
THCCA-Trp	162.6	THCCA-Leu	134.8
THCCA-Ile	147.9	THCCA-Cys	99.8
THCCA-Ala	165.4	THCCA-His	107.5
THCCA-Val	133.2	THCCA-Met	101.9
THCCA-Gln	155.6	THCCA-Pro	87.1

2.2. In vitro anti-platelet aggregation activities of 4a-p

The anti-platelet aggregation activities of **4a-p** were evaluated by *in vitro* assays. In the assays a common procedure was used, while PAF, ADP, AA and TH were used as the aggregators. The final concentration of PAF, ADP, AA and TH were 0.1 μM, 10 μM, 350 μM and 0.1 U/ml, respectively.

The IC_{50} values of **4a-p** against the platelet aggregation induced by PAF and TH are shown in Fig. 2A, B and 2C. The inhibition rates of 100 μM of **4a-p** against the platelet aggregation induced by ADP and AA are listed in Fig. 2D and E.

The data of Fig. 2A–E indicate that **4a-p** feebly inhibit the platelet aggregation induced by PAF, ADP and AA, while strongly inhibit the platelet aggregation induced by TH. For TH induced platelet aggregation the IC_{50} values of **4h,i,j,o** and **4b,c,d,g,k,p** ranged from 0.07 μM to 0.14 μM, and from 0.23 μM to 0.97 μM, respectively. The IC_{50} values of **4a,e, f,l,m,n** against TH induced platelet aggregation ranged from 1.75 μM to 9.56 μM. Thus the assays revealed that the order of the *in vitro* anti-platelet aggregation activity was **4h**,**i**,**j**,**o** > **4b-d**,**g**,**k**,**p** > **4a**,**e**,**f**,**l**,**m**,**n**, and in particular **4j** had the lowest value (0.070 ± 0.003 μM).

2.3. Docking plus in vitro assay for discovering 4j

Table 1 suggests that the order of the DockScores of **4a-p** is **4h**,**i**,**j** > **4b**,**c**,**d**,**g**,**k**,**o**,**p** > **4a**,**e**,**f**,**l**,**m**,**n**. In particular, **4j** gets the highest DockScore (185.7). Fig. 2B and C shows that the order of IC_{50} values of **4a-p** against the platelet aggregation induced by TH is also **4h**,**i**,**j**,**o** > **4b**,**c**,**d**, **g**,**k**,**p** > **4a**,**e**,**f**,**l**,**m**,**n**. In particular, **4j** has the lowest IC_{50} value (0.070 ±

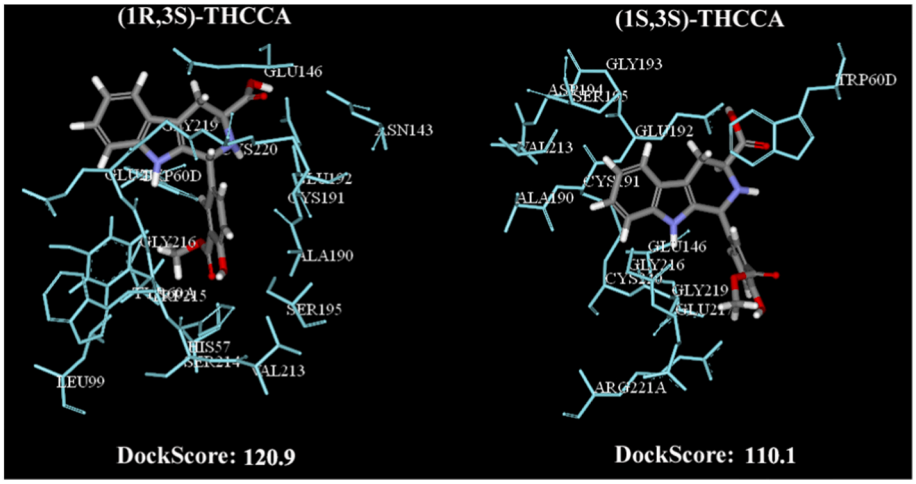
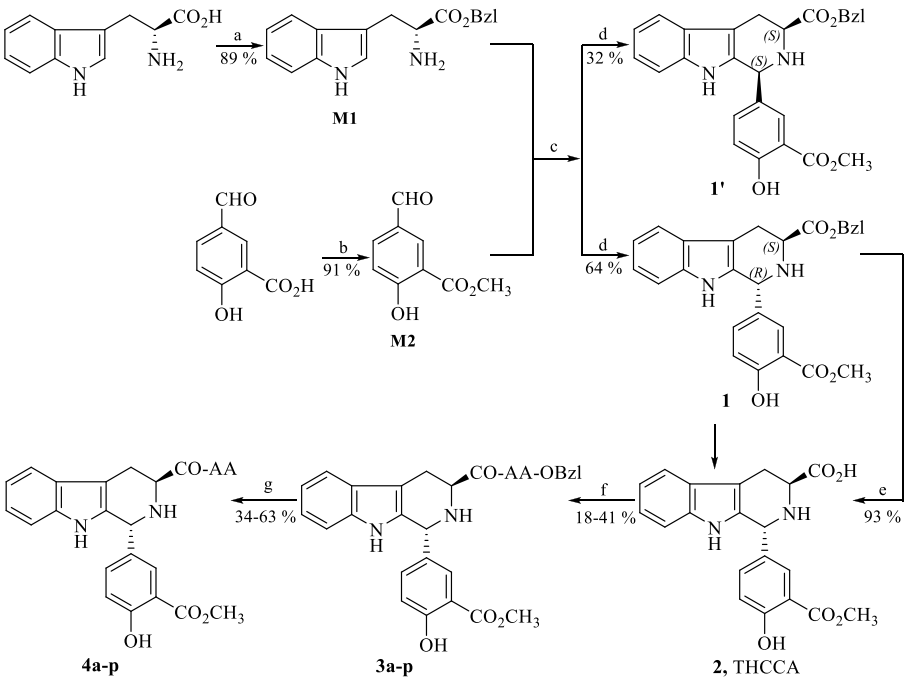


Fig. 1. Stereo views and Dockscores of (1*R*,3*S*)-THCCA and (1*S*,3*S*)-THCCA docked in p pocket of bovine thrombin.



Scheme 1. Reagents and conditions: a) Polyphosphoric acid, benzalcohol, 75 °C; b) Methanol, sulfuric acid; c) Dichloromethane, trifluoroacetic acid, 25 °C; d) silica gel chromatography; e) Pd/C, H_2, tetrahydrofuran/methanol; f) HOBt, DCC, amino acid benzyl ester, 0 °C; g) Pd/C, H_2, methanol. In **3a** and **4a** AA = Gly residue, in **3b** and **4b** AA = *L*-Ala residue, in **3c** and **4c** AA = *L*-Val residue, in **3d** and **4d** AA = *L*-Ile residue, in **3e** and **4e** AA = *L*-Leu residue, in **3f** and **4f** AA = *L*-Phe residue, in **3g** and **4g** AA = *L*-Trp residue, in **3h** and **4h** AA = *L*-Tyr residue, in **3i** and **4i** AA = *L*-Ser residue, in **3j** and **4j** AA = *L*-Asn residue, in **3k** and **4k** AA = *L*-Gln residue, in **3l** and **4l** AA = *L*-Thr residue, in **3m** and **4m** AA = *L*-Asp residue, in **3n** and **4n** AA = *L*-Glu residue, in **3o** and **4o** AA = *L*-Lys residue, in **3p** and **4p** AA = *L*-Arg residue.

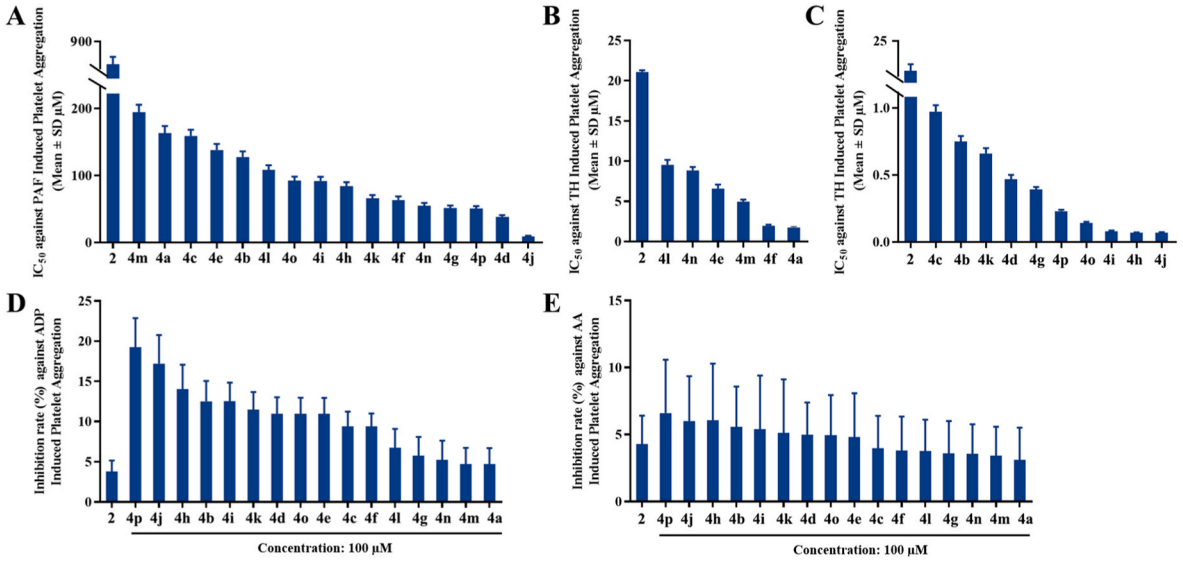


Fig. 2. *In vitro* anti-platelet aggregation activities of **4a-p**. (A) IC_{50} values of **4a-p** against the platelet aggregation induced by PAF, which are higher than 100 μM; (B) IC_{50} values of **4a,e,f,l m,n** against the platelet aggregation induced by TH, which range from 1.75 μM to 9.56 μM; (C) IC_{50} values of **4h,i,j,o** against the platelet aggregation induced by TH, which range from 0.07 μM to 0.14 μM, and IC_{50} values of **4b,c,d,g,k,p** against the platelet aggregation induced by TH, which range from 0.23 μM to 0.97 μM; (D) The inhibition rate of 100 μM of **4a-p** against the platelet aggregation induced by ADP is lower than 20%; (E) The inhibition rate of 100 μM of **4a-p** against the platelet aggregation induced by AA is lower than 10%; n = 6.

0.003 μM). This means that the results of the docking experiments are highly consistent with those of the *in vitro* assays. Therefore a conclusion of **4j** is the selective inhibitor of TH should be credible for both of the docking experiment and the *in vitro* assay.

2.4. *In vivo* anti-thrombotic activities of 4a-p

The anti-thrombotic activities of **4a-p** were evaluated by *in vivo* assay. The activity was assayed on thrombosis model of rats and was represented by thrombus weight. For this purpose the rats were orally administered with NS (negative control) or aspirin at 167 μmol/kg and 16.7 μmol/kg of dosage (positive control) or **2** at 100 nmol/kg of dosage (parent substance control) or each of **4a-p** at 1 nmol/kg of dosage. The thrombus weights of the treated rats were measured and are shown in Fig. 3A–C.

Fig. 3A suggests that the thrombus weights (19.19 mg–20.56 mg) of the rats treated by **4i,o,h,j** at 1 nmol/kg of dosage are significantly lower than that (23.17 mg) of the rats treated by **2** at 10 nmol/kg of dosage. Therefore the anti-thrombosis activity of **4j** has the highest value.

Fig. 3B represents that s the thrombus weights (23.69 mg–25.59 mg) of the rats treated by **4a**,**c**,**e**,**f**,**l**,**m**,**n** at 1 nmol/kg of dosage are equal to

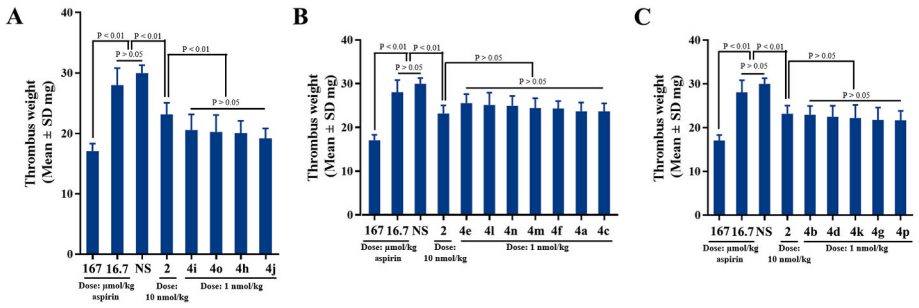


Fig. 3. *In vivo* anti-arterial thrombosis activities of **4a-p**. (A) Thrombus weights (19.19 mg–20.56 mg) of the rats treated by **4i,o,h,j** at 1 nmol/kg of dosage, which are significantly lower than that (23.17 mg) of the rats treated by **2** at 10 nmol/kg of dosage; (B) Thrombus weights (23.69 mg–25.59 mg) of the rats treated by **4a,c,e,f,l,m,n** at 1 nmol/kg of dosage, which are equal to that (23.17 mg) of the rats treated by **2** at 10 nmol/kg of dosage; (C) Thrombus weights (21.66 mg–22.95 mg) of the rats treated by **4b,d,g,k, p** at 1 nmol/kg of dosage, which are significantly lower than that (23.17 mg) of the rats treated by **2** at 10 nmol/kg of dosage; n = 12.

that (23.17 mg) of the rats treated by **2** at 10 nmol/kg of dosage. All the thrombus weights of the rats treated by **4a,c,e,f,l,m,n** show no significant difference.

Fig. 3C denotes that the thrombus weights (21.66 mg–22.95 mg) of the rats treated by **4b,d,g,k,p** at 1 nmol/kg of dosage are also equal to that (23.17 mg) of the rats treated by **2** at 10 nmol/kg of dosage. Again all the thrombus weights of the rats treated by **4b,d,g,k,p** show no significant difference.

2.5. QSAR analysis of the 16 THCCA-amino acids

To explore the contributions of electrostatic, steric and hydrophobic effects to the *in vivo* anti-thrombotic activity of **4a-p** QSAR analysis was performed. Briefly, the molecular field analysis (MFA) model of the Cerius2 was generated by using the CSS alignment. To obtain a consensus alignment, (1*R*,3*S*)-THCCA was selected as the template for superposing **4a-p**. Fig. 4 visualizes aligned **4a-p**. Fig. 4A is the template for superposing **4a-p** with (1*R*,3*S*)-THCCA as the consensus alignment. Fig. 4B represents that near the probes **4j** has an electron donating group and a hydrophilic group, thus is antithrombotic active. Fig. 4C indicates that near the probes **4n** has an electron withdrawing group, thus is antithrombotic inactive.

To superimpose onto (1*R*,3*S*)-THCCA the C-terminal amino acid residue of each structure of **4a-p** has to take individual conformation. G/PLS analysis was carried out for the 16 (1*R*,3*S*)-THCCA-amino acids, the best equation is equation (1), wherein n = 16, $R^2 = 0.958$, r = 0.979 and BS r^2 = 0.942.

Thrombosis inhibition rate = 22.99 + 0.031*"H+/534" + 0.084*"H+/544" + 0.067*"H+/599" - 0.044*"HO-/465" - 0.041*"CH_3/481" - 0.13*"CH_3/692"(1)

As seen, equation (1) contains 3 terms of H+ with the positive coefficient (H+/534, H+/544 and H+/599), 1 term of HO- descriptor with the negative coefficient (HO-/465) and 2 terms of CH_3 descriptor with the negative coefficient (CH_3/481 and CH_3/692). This means that at these positions an electron donating group or a hydrophilic group will positively affect on the antithrombotic activity and vice versa.

2.6. Relationship between docking and in vivo assay of 4j

Table 1 represents that as a member of the DockScores of **4a-p** the DockScore of **4j** is the highest one (185.7). Fig. 3 indicate that as a member of the thrombus weights of the rats treated by **4a-p** at 1 nmol/kg of dosage the thrombus weight of the rats treated by **4j** at 1 nmol/kg of dosage is the lowest one (19.19 mg). A conclusion of **4j** being the selective inhibitor of TH is credible for both the docking experiments and the *in vivo* assays. Therefore **4j** should receive the dose dependent assay *in vivo*.

2.7. Dose dependent action of 4j

To clarify the dose dependent action of **4j** the *in vivo* assays were performed on rat model and the activities were represented by thrombus weight. In order to achieve this aim the rats were orally administered with NS (negative control) or aspirin at 167 μmol/kg and 16.7 μmol/kg of dosage (positive control) or **2** at 100 nmol/kg of dosage (parent substance control) or **4j** at three dosages (1 nmol/kg, 0.01 nmol/kg and 0.001 nmol/kg). The thrombus weights of the treated rats are shown in Fig. 5.

The data in Fig. 5 imply that the thrombus weight of the rats treated by **4j** at 0.001 nmol/kg of dosage is equal to that of the rats treated by NS. Fig. 5 also shows that the thrombus weight of the rats treated by **4j** at 0.001 nmol/kg of dosage is significantly higher than that of the rats treated by **4j** at 0.01 nmol/kg of dosage. Fig. 5 further suggests that the thrombus weight of the rats treated by **4j** at 0.01 nmol/kg of dosage is significantly higher than that of the rats treated by **4j** at 1 nmol/kg of dosage. Thus the assays gave **4j** a dose dependent action *in vivo*, and **4j** should be a promising candidate for developing the selective inhibitor of TH.

2.8. Cell level safety of 4j

The safety at cellular level is critical in the development of **4j**, and the corresponding *in vitro* CCK8 assay was performed. In this profile the normal cells of liver cell (L02), colon mucosa cell (NCM460) and human dermal fibroblasts cell (NHDF) were treated by **4j** at a series of concentrations. Fig. 6 shows that the IC_{50} values of **4j** against the viability of L02, NCM460 and NHDF are higher than 125 μM, which means that the *in vitro* cytotoxicity of **4j** towards mammalian cells is negligible.

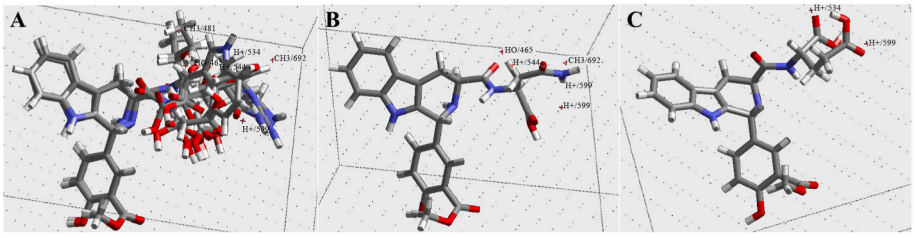


Fig. 4. (A) The template for superposing **4a-p** with (1*R*,3*S*)-THCCA as the consensus alignment; (B) **4j** having an electron donating group and a hydrophilic group near the probes, thus to be antithrombotic active; (C) **4n** having an electron withdrawing group near the probes, thus to be antithrombotic inactive.

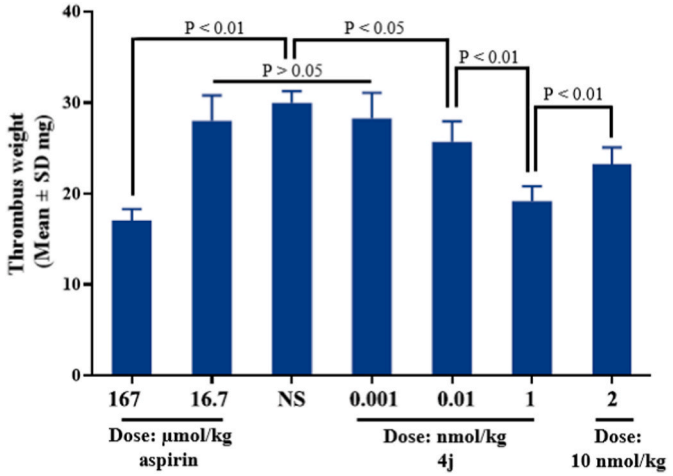


Fig. 5. *In vivo* anti-arterial thrombosis activities of **4j** at 1 nmol/kg, 0.01 nmol/kg and 0.001 nmol/kg of dosage; the thrombus weight of the rats treated by **4j** at 0.001 nmol/kg of dosage is equal to that of the rats treated by NS, and is significantly higher than that of the rats treated by **4j** at 0.01 nmol/kg of dosage; the thrombus weight of the rats treated by **4j** at 0.01 nmol/kg of dosage is significantly higher that of the rats treated by **4j** at 1 nmol/kg of dosage; n = 12.

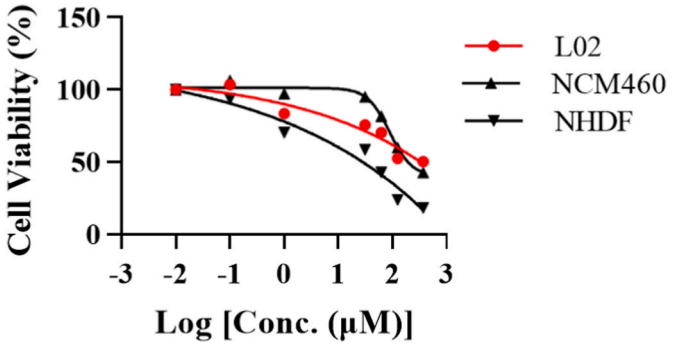


Fig. 6. *In vitro* cytotoxicity of **4j** towards mammalian cells, n = 6.

2.9. Animal level safety of 4j

The safety at animal level is critical in the development of **4j**, and the corresponding *in vivo* assays were performed. In this profile during the anti-thrombotic assay the rats treated by NS, aspirin at 167 μmol/kg of dosage and **4j** at 1 nmol/kg, 0.01 nmol/kg and 0.001 nmol/kg of dosage received tail bleeding time assay. Fig. 7 shows that **4j** does not change the tail bleeding time of the treated rats.

After anti-thrombotic assay the rats treated by NS, aspirin at 167 μmol/kg of dosage and **4j** at 1 nmol/kg, 0.01 nmol/kg and 0.001 nmol/kg of dosage were dissected to examine bleeding condition of the brain, the thoracic cavity and the inferior cavity. It was found that there were no abnormalities in any of these areas of the treated rats.

3. Conclusions

The discovery of the selective inhibitor of TH was addressed by the combination of the virtual screening and the biological assays. The first round of docking experiment led to the selection of (1*R*,3*S*)-THCCA, rather than (1*S*,3*S*)-THCCA, as the leading compound of the selective inhibitor of TH for structural modification. For virtual screening 20 (1*R*,3*S*)-THCCA-amino acids were theoretically designed and the second round of docking experiment was performed, from which 16 (1*R*,3*S*)-THCCA-amino acids were selected for synthesis and biological assay. All

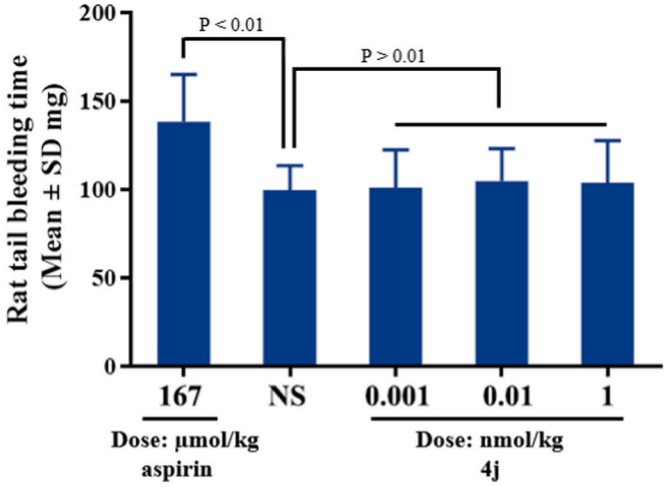


Fig. 7. Tail bleeding time of the anti-thrombotic model rats treated by **4j**, n = 12.

investigations suggest that (1*R*,3*S*)-THCCA-Asn (**4j**) is a promising candidate for developing the selective inhibitor of TH.

4. Experimental section

4.1. General

All chemicals were purchased from Sigma-Aldrich Co (St. Louis, MO, U.S.). and were purified when necessary. Column chromatography was performed with silica gel (200–300 mesh, Qingdao Haiyang Chemical Co., Qingdao, P.R. China). The purity of reaction intermediate was confirmed with TLC (Merck silica gel plates of type 60 F_{254}, 0.25 mm layer thickness, and three different elution procedures), while the purity (more than 96%) of the final product was confirmed with HPLC. In the HPLC analysis, a Waters 2695 HPLC system with a Waters Diode Array Detector was used. The sample was separated on a Symmetry C18 (Waters) reversed-phase column (5 μm, 4.6 × 150 mm). The column thermostat was maintained at 41 °C. 10-μL solution dissolved from the final product in the mobile phase was injected onto the column for analysis. The mobile phase consisted of solvent A (methanol) and solvent B (water). The gradient elution program consisted of 0–5 min 42% A: 58% B, 5–10 min 48% A: 52% B, 10–15 min 58% A: 42% B and 15–20 min 42% A: 58% B, and the flow rate was 0.6 ml/min. After each run, the column was washed with methanol and was equilibrated to initial conditions for 15 min. The DAD detector was set to a scanning range of 200–400 nm. All solvents were distilled and dried before use by the procedure of literature. Melting points were determined in capillary tubes on an electrothermal SM/XMP apparatus without correction. ESI-MS was determined by Micromass Quattro Micro TM API (Waters Co.). Infrared spectra were run on a shimadzu 8010 M spectrophotometer, using the KBr disk method. ^{1}H NMR (500 Hz) and ^{13}C NMR (125 Hz) spectra were recorded on a bruker advance 500 spectrometer, the solvent was $CDCl_3$, TMS was internal standard, and chemical shifts were expressed in ppm. Optical rotations were determined with a jasco P-1020 polarimeter. Biological data was analyzed by using ANOVA test, and $p < 0.05$ was considered statistically significant.

4.2. L-Tryptophan benzyl ester (M1)

A mixture of 15.0 g of polyphosphoric acid, 80 ml of benzalcohol and 10.0 g (49 mmol) of L-tryptophan was stirred at 75 °C for 48 h, and TLC ($CHCl_3$/CH_3OH, 20/1) indicated complete disappearance of L-tryptophan. The reaction mixture was cooled to 0 °C, mixed with 400 ml of anhydrous ether and stirred for 8 h. The formed colorless solids were

collected and suspended with a mixture of 200 ml of ethyl acetate and 10 ml of water. This suspension was rigorously shaken and adjusted to pH 8 with aqueous ammonia. The separated ethyl acetate phase was washed successively with aqueous sodium carbonate (10%) and saturated solution NaCl in water, dried with anhydrous sodium sulfate for 12 h, and evaporated under vacuum. The residue was purified on column chromatography (chloroform/methanol, 30/1) to provide 12.85 g (89%) of L-tryptophan benzyl ester as colorless powders. ESI/MS (*m/e*): 295 $[M + H]^+$.

4.3. Methyl 5-formyl-salicylate (M2)

In a microwave reactor the mixture of 1.66 g (10 mmol) of 5-formyl-salicylic acid, 25 ml of methanol and 0.1 ml of concentrated sulfuric acid was heated at 90 °C for 2 h, and TLC ($CHCl_3/CH_3OH$, 20/1) indicated complete disappearance of 5-formyl- salicylic acid. The reaction mixture was cooled to 25 °C and adjusted to pH 8 with aqueous ammonia, evaporated under vacuum and the residue was purified on column chromatography using chloroform/methanol (40 : 1) to provide 1.64 g (91%) of methyl 5-formylsalicylate. ESI/MS (*m/e*): 181 $[M + H]^+$.

4.4. (1R,3S) and (1S,3S) isomers of 1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9- tetrahydro-β-carboline-3-carboxylic acid benzyl ester (1 and 1')

A mixture of 100 ml of dichloromethane, 10 ml of trifluoroacetic acid, 11.76 g (40 mmol) of L-tryptophan benzyl ester and 7.92 g (44 mmol) of methyl 5-formylsalicylate was stirred at room temperature for 48 h, and TLC ($CHCl_3/CH_3OH$, 20/1) indicated complete disappearance of L-tryptophan benzyl eater. At 0 °C this reaction mixture was adjusted to pH 8 with aqueous ammonia. The separated dichloromethane phase was washed successively with aqueous sodium carbonate (10%) and saturated solution NaCl in water, dried with anhydrous sodium sulfate for 12 h, and evaporated under vacuum. The residue was purified on column chromatography (petroleum/ethyl acetate, 3/1) to provide 6.70 g (32%) of (1*R*,3*S*)- isomer and 12.84 g (64%) of (1*S*,3*S*)-isomer as colorless powders.

(1*R*,3*S*)-isomer (**1**): Mp 81–82 °C. $[\alpha]_D^{25} = -27.2$ (c = 0.65, CH_3OH). IR (KBr): 3348, 3285, 2957, 2897, 1730, 1672, 1489, 1446, 1381, 1306, 1213, 1103, 1012, 842, 799, 750, 702 cm^{-1}. ESI/MS (*m/e*) 457 $[M + H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.56 (s, 1H), 10.50 (s, 1H), 7.71–6.96 (m, 12H), 5.33 (s, 1H), 5.134 (s, 2H), 3.92 (t, *J* = 5.1 Hz, 1H), 3.85 (s, 3H), 3.13 (dd, *J* = 15.3 Hz, *J* = 5.4 Hz, 1H), 3.03 (dd, *J* = 9.6 Hz, *J* = 3.0 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.82, 169.68, 159.80, 136.63, 136.51, 136.36, 134.72, 129.92, 128.81, 128.32, 127.93, 127.01, 121.37, 118.90, 118.18, 117.71, 112.88, 111.51, 106.82, 65.97, 53.57, 52.91, 52.87, 24.99. Anal. Calcd for $C_{27}H_{24}N_2O_5$: C 71.04, H 5.30, N 6.14. Found: C 70.82, H 5.14, N 6.37.

(1*S*,3*S*)-isomer (**1'**): Mp: 82–83 °C; $[\alpha]_D^{25} = -42.7$ (c = 0.50, CH_3OH); IR (KBr): 3058.7, 3031.4, 2949.6, 2898.0, 2843.5, 1732.4, 1681.5, 1489.1, 1455.0, 1440.5, 1302.4, 1268.5, 1212.7, 1142.8, 1089.2, 794.1, 741.1, 695.7 cm^{-1}; ESI/MS (*m/e*): 457 $[M + H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.83 (s, 1H), 10.50 (s, 1H), 7.71–6.96 (m, 12H), 5.42 (s, 1H), 5.18 (s, 2H), 4.07 (t, *J* = 5.1 Hz, 1H), 3.91 (s, 3H), 3.35 (dd, *J* = 15.3 Hz, *J* = 5.4 Hz, 1H), 3.26 (dd, *J* = 9.6 Hz, *J* = 3.0 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.82, 169.68, 159.80, 136.63, 136.51, 136.36, 134.72, 129.92, 128.81, 128.32, 127.93, 127.01, 121.37, 118.90, 118.18, 117.71, 112.88, 111.51, 106.82, 68.6, 61.3, 53.57, 52.91, 52.87, 24.99. Anal. Calcd for $C_{27}H_{24}N_2O_5$: C 71.04, H 5.30, N 6.14. Found: C 70.80, H 5.15, N 6.36.

4.5. (1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboline- 3-carboxylic acid (2)

In the presence of 400 mg of Pd/C, the hydrogenolysis of a solution of 2.0 g (4.4 mmol) of (1*R*,3*S*)-1-(4- hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro- β-carboline-3-carboxylic acid benzyl ester (**1**) proceeded for 8 h and TLC ($CHCl_3/CH_3OH$, 20/1) indicated complete disappearance of **1**. The reaction mixture was evaporated under vacuum and the residue was purified on column chromatography (chloroform/methanol, 40/1) to provide 1.486 g (93%) of the title compound as colorless powders. Mp 100–102 °C. $[\alpha]_D^{25} = -3.5$ (c = 0.55, CH_3OH). IR (KBr): 3391, 3186, 3061, 2953, 2851, 1680, 1593, 1493, 1443, 1379, 1304, 1215, 1090, 839, 797, 743 cm^{-1}. ESI-MS (*m/e*) 365 $[M - H]^-$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.72 (s, 1H), 7.72 (s, 1H), 7.45 (dd, *J* = 17.4 Hz, *J* = 8.1 Hz, 2H), 7.258 (d, *J* = 7.8 Hz, 1H), 7.07 (t, *J* = 7.8 Hz, 1H), 6.99 (d, *J* = 8.7 Hz, 2H), 5.53 (s, 1H), 3.85 (s, 3H), 3.67 (t, *J* = 6.3 Hz, 1H), 3.11 (dd, *J* = 15.3 Hz, *J* = 6.9 Hz, 1H), 2.98 (dd, *J* = 15.3 Hz, *J* = 7.2 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.25, 169.41, 160.11, 136.79, 136.71, 136.64, 130.82, 126.76, 121.69, 119.04, 118.38, 117.81, 113.27, 111.63, 107.75, 107.72, 53.69, 52.67, 48.98, 24.24. Anal. Calcd for $C_{20}H_{18}N_2O_5$: C 65.57, H 4.95, N 7.65. Found: C 65.36, H 4.81, N 7.42.

4.6. General procedure for preparing N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonyl- phenyl)-2,3,4,9-tetrahydro-β-carboline-3-formyl]-L-amino acid benzyl esters (3a-p)

486 mg (3.6 mmol) of HOBt and 768 mg (3.6 mmol) of DCC were added to a solution of 1.10 g (3.0 mmol) of (1R,3S)-1-(4-hydroxy-3-methoxycarbonylphenyl)- 2,3,4,9-tetrahydro-β-carboline-3-carboxylic acid (**2**) in 10 ml of anhydrous THF at 0 °C and stirred for 30 min. To this mixture a solution of 3.6 mmol of L-amino acid benzyl ester in 10 ml of anhydrous THF was added and adjusted to pH 8 with NMM. The reaction mixture was stirred at room temperature for 8 h and TLC ($CHCl_3/CH_3OH$, 20/1) indicated complete disappearance of **2**. The reaction mixture was evaporated under vacuum and the residue was purified on column chromatography (chloroform/methanol, 100/1) to provide **3a-p** as colorless powders.

4.7. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-gly benzyl ester (3a)

Yield: 583 mg (38%). Mp 85–86 °C. $[\alpha]_D^{25} = -25.3$ (c = 0.85, CH_3OH). IR (KBr): 3316, 2930, 2851, 1744, 1672, 1514, 1441, 1310, 1090, 841, 797, 745, 696 cm^{-1}. ESI/MS (*m/e*) 514 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.69 (s, 1H), 8.37 (s, 1H), 7.64–6.97 (m, 12H), 5.25 (s, 1H), 5.13 (s, 2H), 3.94 (s, 2H), 3.84 (s, 3H), 3.55 (m, 1H), 2.98 (dd, *J* = 15.3 Hz, *J* = 4.8 Hz, 1H), 2.82 (dd, *J* = 15.3 Hz, *J* = 8.4 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.66, 170.26, 169.71, 159.71, 136.58, 136.43, 136.34, 134.64, 134.49, 134.45, 129.84, 127.16, 121.40, 118.85, 118.15, 117.70, 112.79, 111.51, 108.32, 66.32, 53.38, 52.89, 52.26, 33.78; 24.97. Anal. Calcd for $C_{29}H_{27}N_3O_6$: C 67.83, H 5.30, N 8.18. Found: C 67.60, H 5.14, N 7.94.

4.8. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-ala benzyl ester (3b)

Yield: 650 mg (41%). Mp 79–80 °C. $[\alpha]_D^{25} = -32.4$ (c = 0.45, CH_3OH). IR (KBr): 3374, 2953, 2897, 1740, 1676, 1517, 1452, 1306, 1211, 1090, 844, 797, 741, 696 cm^{-1}. ESI/MS (*m/e*) 528 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.75 (s, 1H), 8.32 (d, *J* = 6.9 Hz, 1H), 7.61–6.97 (m, 12H), 5.22 (s, 1H), 5.13 (dd., *J* = 15.3 Hz, *J* = 12.6 Hz, 2H), 4.36 (d, *J* = 7.2 Hz, 1H), 3.83 (s, 3H), 3.51 (dd, *J* = 8.7 Hz, *J* = 4.5 Hz, 1H), 2.94 (dd, *J* = 15.3 Hz, *J* = 4.5 Hz, 1H), 2.72 (dd, *J* = 15.3 Hz, *J* = 9.3 Hz, 1H), 1.33 (d, *J* = 7.2 Hz, 3H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 176.39, 173.29, 173.21, 172.84, 136.72, 136.55, 134.66, 134.58, 134.43, 128.44, 127.16, 127.12, 126.88, 121.41, 118.88, 118.08, 111.58, 111.53, 108.28, 65.96, 53.28, 51.76, 49.93, 48.08, 47.98, 25.47, 20.85, 17.34. Anal. Calcd for $C_{30}H_{29}N_3O_6$: C 68.30, H 5.54, N 7.96. Found: C 68.11, H 5.40, N 7.73.

4.9. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-val benzyl ester (3c)

Yield: 352 mg (21%). Mp 78–80 °C. $[\alpha]_D^{25} = -68.4$ (c = 0.55, CH_3OH). IR (KBr): 3367, 3287, 2961, 2932, 1738, 1676, 1516, 1447, 1307, 1211, 1090, 839, 795, 744, 698 cm^{-1}. ESI/MS (*m/e*) 556 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.74 (s, 1H), 10.53 (s, 1H), 8.10 (d, *J* = 8.1 Hz, 1H), 7.67–6.97 (m, 12H), 5.19 (s, 1H), 5.14 (dd., *J* = 20.4 Hz, *J* = 12.6 Hz, 2H), 4.23 (d, *J* = 5.7 Hz, 1H), 3.82 (s, 3H), 3.57 (dd, *J* = 8.1 Hz, *J* = 5.1 Hz, 1H), 2.94 (dd, *J* = 15.3 Hz, *J* = 4.5 Hz, 1H), 2.71 (dd, *J* = 15.3 Hz, *J* = 8.7 Hz, 1H), 2.15–2.06 (m, 1H), 0.88 (d, *J* = 6.3 Hz, 6H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.16, 171.70, 169.68, 159.62, 136.38, 136.32, 134.69, 134.54, 134.48, 129.67, 128.59, 127.12, 121.39, 118.85, 118.09, 117.68, 112.81, 111.51, 108.25, 72.70, 60.63, 57.73, 53.29, 51.98, 30.35, 25.79, 25.28, 24.91. Anal. Calcd for $C_{32}H_{33}N_3O_6$: C 69.17, H 5.99, N 7.56. Found: C 69.00, H 5.83, N 7.32.

4.10. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-ile benzyl ester (3d)

Yield: 352 mg (23%). Mp 77–79 °C. $[\alpha]_D^{25} = -36.2$ (c = 0.6, CH_3OH). IR (KBr): 3385, 3291, 2965, 2878, 1736, 1676, 1508, 1441, 1301, 1209, 1090, 843, 797, 743, 696 cm^{-1}. ESI/MS (*m/e*) 570 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.74 (s, 1H), 10.52 (s, 1H), 8.11 (d, *J* = 6.6 Hz, 1H), 7.63–6.97 (m, 12H), 5.20 (s, 1H), 5.14 (dd., *J* = 21.3 Hz, *J* = 12.0 Hz, 2H), 4.30 (s, 1H), 3.83 (s, 3H), 3.57 (s, 1H), 2.95 (d, *J* = 12.3 Hz, 1H), 2.75–2.68 (m, 1H), 1.84 (s, 1H), 1.39 (s, 1H), 1.15 (dd, *J* = 34.8 Hz, *J* = 7.2 Hz, 1H), 0.84 (s, 6H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.53, 171.69, 169.68, 159.66, 136.39, 136.31, 134.52, 134.49, 129.71, 128.59, 128.53, 127.12, 121.39, 118.86, 118.10, 117.70, 112.80, 111.52, 108.27, 66.40, 56.79, 56.70, 53.32, 52.88, 51.97, 36.81, 25.31, 15.93, 11.61. Anal. Calcd for $C_{33}H_{35}N_3O_6$: C 69.58, H 6.19, N 7.38. Found: C 69.79, H 6.34, N 7.15.

4.11. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-leu benzyl ester (3e)

Yield: 405 mg (24%). Mp 74–75 °C. $[\alpha]_D^{25} = -61.1$ (c = 0.6, CH_3OH). IR (KBr): 3377, 3277, 2957, 2872, 1740, 1676, 1616, 1510, 1456, 1213, 1090, 847, 797, 743, 696 cm^{-1}. ESI/MS (*m/e*) 570 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.71 (s, 1H), 10.57 (s, 1H), 8.25 (d, *J* = 6.9 Hz, 1H), 7.62–6.99 (m, 12H), 5.19 (s, 1H), 5.12 (s, 1H), 4.35 (s, 1H), 3.83 (s, 3H), 3.55 (s, 1H), 2.94 (d, *J* = 11.7 Hz, 1H), 2.76 (d, *J* = 9.3 Hz, 1H), 1.63–1.60 (m, 3H), 0.87 (d, *J* = 9.6 Hz, 6H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.50, 172.72, 169.68, 159.67, 136.40, 134.56, 134.48, 129.75, 128.89, 128.53, 128.31, 127.12, 121.37, 118.85, 118.08, 117.71, 112.79, 111.52, 108.15, 66.41, 53.30, 52.88, 52.06, 50.76, 39.12, 25.27, 24.75, 23.17, 21.81. Anal. Calcd for $C_{33}H_{35}N_3O_6$: C 69.58, H 6.19, N 7.38. Found: C 69.37, H 6.02, N 7.16.

4.12. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-phe benzyl ester (3f)

Yield: 450 mg (25%). Mp 79–80 °C. $[\alpha]_D^{25} = -25.5$ (c = 0.65, CH_3OH). IR (KBr): 3366, 2953, 2893, 1740, 1672, 1492, 1440, 1301, 1213, 1090, 843, 797, 741, 700 cm^{-1}. ESI/MS (*m/e*) 604 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.64 (s, 1H), 8.30 (d, *J* = 7.5 Hz, 1H), 7.59–6.96 (m, 17H), 5.10 (s, 1H), 5.06 (s, 2H), 4.56 (t, *J* = 6.6 Hz, 1H), 3.85 (s, 3H), 3.52 (t, *J* = 6.3 Hz, 1H), 3.13–2.98 (m, 2H), 2.90 (dd, *J* = 15.0 Hz, *J* = 4.8 Hz, 1H), 2.76 (dd, *J* = 15.3 Hz, *J* = 7.8 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.27, 171.72, 169.70, 159.71, 137.42, 136.40, 136.32, 136.20, 134.42, 134.39, 129.77, 129.61, 128.84, 128.76, 128.51, 128.35, 127.09, 121.37, 118.84, 118.09, 117.69, 112.77, 111.52, 108.10, 66.52, 53.90, 53.23, 52.88, 52.35, 36.90, 24.70. Anal. Calcd for $C_{36}H_{33}N_3O_6$: C 71.63, H 5.51, N 6.96. Found: C 71.85, H 5.66, N 6.73.

4.13. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-trp benzyl ester (3g)

Yield: 785 mg (41%). Mp 96–97 °C. $[\alpha]_D^{25} = -17.9$ (c = 0.7, CH_3OH). IR (KBr): 3397, 3292, 2953, 2849, 1736, 1672, 1618, 1510, 1456, 1301, 1213, 1090, 845, 797, 743, 696 cm^{-1}. ESI/MS (*m/e*) 643 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.91 (s, 1H), 10.70 (s, 1H), 10.52 (s, 1H), 8.25 (d, *J* = 6.9 Hz, 1H), 7.55–7.00 (m, 17H), 5.12 (s, 1H), 5.05 (s, 2H), 4.61 (t, *J* = 6.6 Hz, 1H), 3.81 (s, 3H), 3.57 (s, 1H), 3.21 (s, 2H), 2.94 (d, *J* = 11.7Hz, 1H), 2.74 (d, *J* = 8.7 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6): δ/ppm = 174.51, 171.65, 169.42, 159.06, 136.64, 136.45, 135.67, 135.15, 134.82, 133.09, 128.90, 128.64, 128.44, 128.24, 127.36, 121.91, 121.76, 119.26, 119.10, 118.71, 118.15, 118.05, 113.45, 111.41, 111.30, 109.82. Anal. Calcd for $C_{38}H_{34}N_4O_6$: C 71.01, H 5.33, N 8.72. Found: C 70.78, H 5.17, N 8.95.

4.14. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-tyr benzyl ester (3h)

Yield: 650 mg (35%). Mp 88–89 °C. $[\alpha]_D^{25} = -34.4$ (c = 0.4, CH_3OH: THF = 1 : 2). IR (KBr): 3335, 2955, 2885, 1744, 1672, 1597, 1514, 1441, 1304, 1213, 1092, 833, 797, 743, 700 cm^{-1}. ESI/MS (*m/e*) 620 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.68 (s, 1H), 8.22 (d, *J* = 7.2 Hz, 1H), 7.60–6.63 (m, 16H), 5.12 (s, 1H), 5.08 (s, 2H), 4.46 (t, *J* = 6.3Hz, 1H), 3.84 (s, 3H), 3.50 (t, *J* = 6.3Hz, 1H), 2.94 (s, 2H), 2.88 (d, *J* = 5.1Hz, 1H), 2.74 (dd, *J* = 15.3 Hz, *J* = 8.1 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.31, 171.82, 169.71, 159.77, 156.54, 136.55, 136.40, 136.35, 136.22, 135.67, 134.60, 134.45, 134.32, 130.57, 128.83, 128.50, 128.35, 127.29, 127.16, 127.12, 121.40, 118.86, 118.10, 117.73, 115.58, 112.79, 111.53, 108.24, 108.21, 66.44, 54.32, 53.31, 52.88, 52.28, 36.27, 24.78. Anal. Calcd for $C_{36}H_{33}N_3O_7$: C 69.78, H 5.37, N 6.78. Found: C 69.56, H 5.12, N 6.57.

4.15. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-ser benzyl ester (3i)

Yield: 450 mg (28%). Mp 85–86 °C. $[\alpha]_D^{25} = -14.3$ (c = 0.95, CH_3OH). IR (KBr): 3379, 3277, 2953, 2851, 1740, 1672, 1514, 1441, 1306, 1213, 1090, 845, 797, 743, 696 cm^{-1}. ESI/MS (*m/e*) 544 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.73 (s, 1H), 10.52 (s, 1H), 8.24 (d, *J* = 7.8 Hz, 1H), 7.63–6.97 (m, 12H), 5.23 (s, 1H), 5.15 (s, 2H), 4.44 (s, 1H), 3.83 (s, 3H), 3.72 (dd, *J* = 20.7 Hz, *J* = 4.5 Hz, 1H), 3.57 (dd, *J* = 8.1 Hz, *J* = 4.8 Hz, 1H), 2.98 (dd, *J* = 15.6 Hz, *J* = 4.5 Hz, 1H), 2.75 (dd, *J* = 15.3Hz, *J* = 9.3Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.37, 170.82, 169.71, 159.67, 136.58, 136.45, 136.40, 134.59, 134.44, 134.37, 129.78, 128.44, 127.18, 127.14, 121.44, 118.88, 118.14, 117.70, 112.79, 111.58, 108.48, 66.45, 61.53, 55.03, 53.42, 52.89, 51.95, 25.07. Anal. Calcd for $C_{30}H_{29}N_3O_7$: C 66.29, H 5.38, N 7.73. Found: C 66.50, H 5.54, N 7.96.

4.16. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-asn benzyl ester (3j)

Yield: 685 mg (40%). Mp 84–85 °C. $[\alpha]_D^{25} = -75.5$ (c = 0.65, CH_3OH). IR (KBr): 3370, 2953, 2849, 1740, 1674, 1510, 1441, 1310, 1209, 1090, 845, 797, 746, 696 cm^{-1}. ESI/MS (*m/e*) 571 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.69 (s, 1H), 10.53 (s, 1H), 8.41 (d, *J* = 7.8 Hz, 1H), 7.62–6.97 (m, 14H), 5.19 (s, 1H), 5.11 (s, 2H), 4.69 (s, 1H), 3.83 (s, 3H), 3.54 (s, 1H), 2.95 (dd, *J* = 15.0 Hz, *J* = 4.5 Hz, 1H), 2.79 (dd, *J* = 15.0Hz, *J* = 8.4Hz, 1H), 2.64 (d, *J* = 4.8 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.08, 173.01, 171.64, 169.73, 159.69, 136.55, 136.43, 134.62, 134.47, 134.34, 129.72, 128.82, 128.38, 128.05, 127.20, 127.16, 121.41, 118.86, 118.12, 117.71, 112.78, 111.54, 108.33, 66.46, 55.48, 52.89, 52.20, 49.06, 36.90, 24.78. Anal.

Calcd for $C_{31}H_{30}N_4O_7$: C 65.25, H 5.30, N 9.82. Found: C 65.46, H 5.45, N 10.04.

4.17. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-gln benzyl ester (3k)

Yield: 300 mg (17%). Mp 111–112 °C. $[\alpha]_D^{25} = -33.6$ (c = 1.15, CH_3OH). IR (KBr): 3277, 3196, 2953, 2927, 1736, 1676, 1616, 1492, 1440, 1308, 1209, 1090, 843, 795, 743, 698 cm^{-1}. ESI/MS (*m/e*) 585 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.72 (s, 1H), 10.50 (s, 1H), 8.29 (d, J = 7.2 Hz, 1H), 7.61–6.80 (m, 14H), 5.22 (s, 1H), 5.12 (s, 2H), 4.31 (dd, J = 8.7 Hz, J = 4.8 Hz, 1H), 3.83 (s, 3H), 3.54–3.49 (m, 1H), 2.95 (dd, J = 15.3 Hz, J = 4.5 Hz, 1H), 2.71 (dd, J = 15.3Hz, J = 9.0Hz, 1H), 2.16 (t, J = 7.2 Hz, 2H), 2.07–1.96 (m, 1H), 1.92–1.87 (m, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.81, 173.52, 172.12, 169.69, 159.69, 136.38, 134.55, 134.44, 129.72, 128.88, 128.51, 128.29, 127.10, 121.42, 118.88, 118.09, 117.67, 112.76, 111.53, 108.25, 66.44, 53.29, 52.89, 52.15, 51.87, 31.58, 26.86, 25.39. Anal. Calcd for $C_{32}H_{32}N_4O_7$: C 65.74, H 5.52, N 9.58. Found: C 65.53, H 5.36, N 9.35.

4.18. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-thr benzyl ester (3l)

Yield: 645 mg (39%). Mp 85–86 °C. $[\alpha]_D^{25} = -22.7$ (c = 1.05, CH_3OH). IR (KBr): 3397, 3302, 2947, 2850, 1741, 1676, 1492, 1447, 1308, 1211, 1090, 841, 797, 745, 696 cm^{-1}. ESI/MS (*m/e*) 558 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.72 (s, 1H), 8.04 (d, J = 8.7 Hz, 1H), 7.65–6.97 (m, 12H), 5.24 (s, 1H), 5.14 (s, 2H), 4.35 (d, J = 3.0 Hz, 1H), 4.21 (t, J = 3.0 Hz, 1H), 3.83 (s, 3H), 3.61 (t, J = 6.3 Hz, 1H), 3.01 (dd, J = 15.3 Hz, J = 4.8 Hz, 1H), 2.80 (dd, J = 15.3 Hz, J = 8.4 Hz, 1H), 1.11 (d, J = 6.3 Hz, 3H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.56, 170.95, 169.70, 159.73, 136.59, 136.44, 136.40, 134.70, 134.55, 134.23, 129.74, 128.84, 128.43, 128.13, 127.21, 127.17, 121.44, 118.87, 118.18, 117.76, 112.84, 111.52, 108.50, 66.43, 58.13, 55.35, 53.52, 52.89, 52.32, 24.81, 20.80. Anal. Calcd for $C_{31}H_{31}N_3O_7$: C 66.77, H 5.60, N 7.54. Found: C 66.56, H 5.44, N 7.75.

4.19. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-asp dibenzyl ester (3 m)

Yield: 350 mg (18%). Mp 75–76 °C. $[\alpha]_D^{25} = -16.6$ (c = 0.9, CH_3OH). IR (KBr): 3377, 2953, 2926, 1736, 1676, 1618, 1496, 1440, 1213, 1090, 843, 797, 743, 696 cm^{-1}. ESI/MS (*m/e*) 662 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.68 (s, 1H), 10.51 (s, 1H), 8.48 (d, J = 7.8 Hz, 1H), 7.63–6.99 (m, 17H), 5.20 (s, 1H), 5.10 (s, 2H), 5.08 (s, 2H), 4.79 (t, J = 5.7 Hz, 1H), 3.82 (s, 3H), 3.54 (t, J = 6.3 Hz, 1H), 3.00–2.83 (m, 3H), 2.77 (dd, J = 15.3 Hz, J = 7.8 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.19, 170.92, 170.49, 169.69, 159.72, 136.58, 136.42, 136.24, 136.16, 134.58, 134.41, 129.80, 128.84, 128.49, 128.37, 128.22, 127.12, 121.40, 118.86, 118.10, 117.72, 112.81, 111.53, 108.20, 66.81, 66.38, 52.31, 52.87, 52.16, 48.97, 36.15, 24.99. Anal. Calcd for $C_{38}H_{35}N_3O_8$: C 68.97, H 5.33, N 6.35. Found: C 69.16, H 5.48, N 6.11.

4.20. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-glu dibenzyl ester (3n)

Yield: 450 mg (22%). Mp 101–102 °C. $[\alpha]_D^{25} = -21.1$ (c = 0.8, CH_3OH). IR (KBr): 3370, 2953, 2926, 2849, 1736, 1676, 1618, 1492, 1449, 1211, 1090, 847, 797, 743, 696 cm^{-1}. ESI/MS (*m/e*) 676 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.71 (s, 1H), 10.49 (s, 1H), 8.28 (d, J = 7.8 Hz, 1H), 7.63–6.95 (m, 17H), 5.21 (s, 1H), 5.12 (s, 2H), 5.07 (s, 2H), 4.38 (s, 1H), 3.81 (s, 3H), 3.52 (s, 1H), 2.94 (d, J = 12.6 Hz, 1H), 2.72 (dd, J = 15.3 Hz, J = 8.7 Hz, 1H), 2.46 (s, 2H), 2.09 (d, J = 6.9 Hz, 1H), 1.97 (d, J = 6.9 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.58, 172.50, 171.86, 169.69, 159.61, 136.56, 136.33, 134.63, 134.48, 129.70, 128.87, 128.53, 128.46, 128.37, 128.32, 127.14, 121.40, 118.87, 118.08, 117.64, 112.75, 111.53, 108.18, 66.55, 66.01, 53.30, 52.86, 51.96, 51.61, 30.31, 26.37, 25.33. Anal. Calcd for $C_{39}H_{37}N_3O_8$: C 69.32, H 5.52, N 6.22. Found: C 69.51, H 5.36, N 6.00.

4.21. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-lys(Z) benzyl ester (3°)

Yield: 450 mg (40%). Mp 84–85 °C. $[\alpha]_D^{25} = -45.9$ (c = 0.55, CH_3OH). IR (KBr): 3300, 2951, 2864, 1701, 1676, 1514, 1440, 1213, 1090, 843, 797, 741, 696 cm^{-1}. ESI/MS (*m/e*): 719 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.71 (s, 1H), 10.51 (s, 1H), 8.31 (s, 1H), 8.22 (d, J = 7.2 Hz, 1H), 7.62–6.99 (m, 17H), 5.21 (s, 1H), 5.13 (s, 2H), 4.99 (s, 2H), 4.28 (t, J = 6.9 Hz, 1H), 3.82 (s, 3H), 3.54 (dd, J = 8.1 Hz, J = 4.8 Hz, 1H), 2.98 (s, 2H), 2.73 (dd, J = 13.5 Hz, J = 8.1 Hz, 1H), 1.71 (q, J = 7.5 Hz, 2H), 1.35 (d, J = 24.6 Hz, J = 6.9 Hz, 4H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.50, 172.34, 169.70, 159.63, 156.58, 137.71, 136.40, 134.58, 134.46, 129.73, 128.87, 128.80, 128.52, 128.32, 128.19, 128.17, 127.12, 121.40, 118.87, 118.10, 117.67, 112.73, 111.53, 108.23, 66.40, 65.60, 53.30, 52.87, 52.41, 51.94, 30.89, 29.47, 25.34, 23.11. Anal. Calcd for $C_{41}H_{42}N_4O_8$: C 68.51, H 5.89, N 7.79. Found: C 68.70, H 6.05, N 7.58.

4.22. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-N^G-NO_2-L-arg benzyl ester (3p)

Yield: 675 mg (34%). Mp 117–118 °C. $[\alpha]_D^{25} = -56.5$ (c = 0.45, CH_3OH). IR (KBr): 3275, 3237, 2953, 2895, 1740, 1673, 1518, 1493, 1441, 1213, 1090, 845, 797, 743, 696 cm^{-1}. ESI/MS (*m/e*) 658 $[M+H]^+$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.72 (s, 1H), 10.50 (s, 1H), 8.31 (s, 1H), 8.28 (d, J = 7.2 Hz, 1H), 7.61–6.97 (m, 12H), 5.21 (s, 1H), 5.13 (s, 2H), 4.33 (s, 1H), 3.83 (s, 3H), 3.54 (t, J = 4.8 Hz, 1H), 3.16 (s, 2H), 2.95 (dd, J = 15.0 Hz, J = 3.9 Hz, 1H), 2.73 (dd, J = 15.0 Hz, J = 9.0 Hz, 1H), 1.82–1.63 (m, 2H), 1.54 (s, 2H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.62, 173.55, 172.17, 169.69, 159.61, 136.55, 136.40, 136.33, 134.58, 134.44, 129.71, 128.88, 128.54, 128.32, 127.10, 121.42, 118.88, 118.09, 117.68, 112.79, 111.54, 108.21, 79.62, 66.51, 53.32, 52.89, 52.24, 52.14, 51.96, 28.42, 25.39. Anal. Calcd for $C_{33}H_{35}N_7O_8$: C 60.27, H 5.36, N 14.91. Found: C 60.50, H 5.20, N 14.69.

4.23. General procedure for preparing N-[(1R,3S)-1-(4-hydroxy-3-methoxycarbonyl- phenyl)-2,3,4,9-tetrahydro-β-carboline-3-formyl]-L-amino acids (4a-p)

In the presence of 40 mg of Pd/C, the hydrogenolysis of a solution of 200 mg of N-[(1*R*,3*S*)-1-(4-hydroxy- 3-methoxycarbonylphenyl)-2,3,4,9-tetrahydro-β-car-boline-3-formyl]-L-amino acid benzyl ester (**3a-p**) was proceeded for 8 h and TLC ($CHCl_3/CH_3OH$, 20/1) indicated complete disappearance of the benzyl ester (**3a-p**). The reaction mixture was evaporated under vacuum and the residue was purified on column chromatography (chloroform/methanol, 30/1) to provide the title compound as colorless powders.

4.24. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-gly (4a)

Yield: 85 mg (52%). Mp 96–97 °C. $[\alpha]_D^{25} = -93.3$ (c = 0.45, CH_3OH). IR (KBr): 3375, 3217, 3063, 2953, 2859, 1678, 1593, 1493, 1443, 1308, 1215, 1092, 841, 797, 745 cm^{-1}. ESI-MS (*m/e*) 422 $[M-H]^-$; ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.72 (s, 1H), 8.18 (s, 1H), 7.64 (s, 1H), 7.46 (s, 2H), 7.26 (s, 1H), 7.01 (s, 3H), 5.28 (s, 1H), 4.06 (s, 1H), 3.84 (s, 3H), 3.79 (s, 2H), 3.01 (d, J = 12.6 Hz, 1H), 2.83 (d, J = 12.6 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.08, 171.65, 169.64, 159.75, 136.52, 136.43, 134.14, 133.94, 130.02, 127.08, 121.46, 118.89,

118.20, 117.73, 112.87, 111.53, 108.33, 65.39, 53.43, 52.90, 52.10, 24.86. Anal. Calcd for $C_{22}H_{21}N_3O_6$: C 62.41, H 5.00, N 9.92. Found: C 62.20, H 4.84, N 9.70. HPLC purity: 92.8%.

4.25. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-ala (4b)

Yield: 86 mg (52%). Mp 120–122 °C. $[\alpha]_D^{25}$ = −70.7 (c = 0.45, CH_3OH). IR (KBr): 3379, 3221, 3059, 2953, 2882, 1678, 1595, 1493, 1449, 1308, 1217, 1092, 841, 797, 745 cm^{-1}. ESI-MS (*m/e*) 436 $[M-H]^-$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.75 (s, 1H), 8.17 (d, *J* = 7.2 Hz, 1H), 7.62 (s, 1H), 7.42 (dd, *J* = 12.9 Hz, *J* = 7.5 Hz, 2H), 7.28 (d, *J* = 7.5 Hz, 1H), 7.06 (t, *J* = 7.5 Hz, 1H), 6.99 (d, *J* = 8.4 Hz, 2H), 5.26 (s, 1H), 4.21 (d, *J* = 6.9 Hz, 1H), 3.83 (s, 3H), 3.54 (dd, *J* = 8.4 Hz, *J* = 4.5 Hz, 1H), 3.01 (dd, *J* = 15.3 Hz, *J* = 4.5 Hz, 1H), 2.76 (dd, *J* = 15.0 Hz, *J* = 9.0 Hz, 1H), 1.29 (d, *J* = 6.9 Hz, 3H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 174.37, 172.49, 169.60, 159.70, 136.52, 136.46, 134.14, 133.85, 129.99, 127.03, 121.49, 118.93, 118.16, 117.70, 112.93, 111.55, 108.27, 53.40, 52.89, 51.76, 47.83, 25.14, 17.78. Anal. Calcd for $C_{23}H_{23}N_3O_6$: C 63.15, H 5.30, N 9.61. Found: C 63.37, H 5.45, N 9.84. HPLC purity: 91.7%.

4.26. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-val (4c)

Yield: 95 mg (57%). Mp 137–138 °C. $[\alpha]_D^{25}$ = −50.9 (c = 0.5, CH_3OH). IR (KBr): 3377, 3291, 3063, 2963, 2876, 1678, 1593, 1493, 1443, 1308, 1215, 1090, 839, 797, 745 cm^{-1}. ESI-MS (*m/e*) 464 $[M-H]^-$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.74 (s, 1H), 10.51 (s, 1H), 7.94 (d, *J* = 8.1 Hz, 1H), 7.65 (s, 1H), 7.45 (t, *J* = 8.4 Hz, 2H), 7.28 (d, *J* = 7.5 Hz, 1H), 7.05 (t, *J* = 7.8 Hz, 1H), 6.99 (d, *J* = 7.2 Hz, 2H), 5.25 (s, 1H), 4.16 (t, *J* = 6.9 Hz, 1H), 3.84 (s, 3H), 3.61 (s, 1H), 3.02 (dd, *J* = 12.6 Hz, *J* = 7.2 Hz, 1H), 2.78 (dd, *J* = 14.7 Hz, *J* = 8.7 Hz, 1H), 2.08 (d, *J* = 6.0 Hz, 1H), 0.90 (d, *J* = 5.4 Hz, 6H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.26, 173.01, 169.64, 159.82, 136.62, 136.34, 134.35, 133.86, 129.91, 127.13, 121.46, 118.90, 118.17, 117.78, 113.01, 111.59, 108.32, 57.47, 53.58, 52.88, 52.23, 30.49, 25.08, 19.59, 18.46. Anal. Calcd for $C_{25}H_{27}N_3O_6$: C 64.50, H 5.85, N 9.03. Found: C 64.72, H 6.01, N 9.27. HPLC purity: 91.8%.

4.27. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-ile (4d)

Yield: 115 mg (68%). Mp 108–109 °C. $[\alpha]_D^{25}$ = −43.5 (c = 0.55, CH_3OH). IR (KBr): 3383, 3202, 3061, 2965, 2876, 1678, 1593, 1493, 1443, 1383, 1306, 1215, 1090, 841, 797, 745 cm^{-1}. ESI-MS (*m/e*) 478 $[M-H]^-$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.75 (s, 1H), 8.21 (s, 1H), 7.95 (d, *J* = 7.2 Hz, 1H), 7.63 (s, 1H), 7.45 (t, *J* = 8.4 Hz, 2H), 7.27 (d, *J* = 7.5 Hz, 1H), 7.05 (t, *J* = 7.5 Hz, 1H), 6.99 (d, *J* = 7.5 Hz, 2H), 5.21 (s, 1H), 4.17 (d, *J* = 4.5 Hz, 1H), 3.83 (s, 3H), 3.57 (s, 1H), 3.00 (d, *J* = 11.4 Hz, 1H), 2.76 (dd, *J* = 14.7 Hz, *J* = 8.7 Hz, 1H), 1.81 (s, 1H), 1.24–1.17 (m, *J* = 5.4 Hz, 1H), 0.87 (d, *J* = 5.4 Hz, 6H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.30, 172.88, 169.61, 159.68, 136.41, 136.33, 134.34, 134.03, 129.85, 127.10, 121.43, 118.88, 118.16, 117.73, 112.91, 111.53, 108.34, 56.57, 53.47, 52.89, 52.15, 37.07, 25.25, 25.02, 16.05, 11.85. Anal. Calcd for $C_{26}H_{29}N_3O_6$: C 65.12, H 6.10, N 8.76. Found: C 65.35, H 6.26, N 8.53. HPLC purity: 98.9%.

4.28. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-leu (4e)

Yield: 105 mg (62%). Mp 115–116 °C. $[\alpha]_D^{25}$ = −49.5 (c = 0.55, CH_3OH). IR (KBr): 3368, 3217, 3061, 2957, 2872, 1680, 1591, 1493, 1443, 1310, 1213, 1092, 841, 797, 745 cm^{-1}. ESI-MS (*m/e*) 478 $[M-H]^-$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.71 (s, 1H), 8.20 (s, 1H), 8.07 (d, *J* = 7.8 Hz, 1H), 7.63 (s, 1H), 7.44 (t, *J* = 8.4 Hz, 2H), 7.27 (d, *J* = 7.8 Hz, 1H), 7.06 (t, *J* = 7.5 Hz, 1H), 6.99 (d, *J* = 8.4 Hz, 2H), 5.21 (s, 1H), 4.23 (d, *J* = 7.8 Hz, 1H), 3.84 (s, 3H), 3.56 (t, *J* = 5.1 Hz, 1H), 3.00 (dd, *J* = 15.3 Hz, *J* = 4.5 Hz, 1H), 2.78 (dd, *J* = 15.0Hz, *J* = 9.0 Hz, 1H), 1.63–1.51 (m, *J* = 19.5 Hz, *J* = 9.0 Hz, *J* = 6.9 Hz, 3), 0.87 (dd, *J* = 13.5 Hz, *J* = 6.0 Hz, 6); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 174.41, 172.95, 169.64, 163.81, 159.72, 136.44, 136.35, 134.34, 134.19, 129.88, 127.12, 121.40, 118.87, 118.13, 117.73, 112.89, 111.52, 108.23, 53.46, 52.87, 52.21, 50.66, 25.02, 24.83, 23.30, 21.94. Anal. Calcd for $C_{26}H_{29}N_3O_6$: C 65.12, H 6.10, N 8.76. Found: C 65.35, H 6.26, N 8.52. HPLC purity: 98.3%.

4.29. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-phe (4f)

Yield: 98 mg (58%). Mp 115–116 °C. $[\alpha]_D^{25}$ = −54.5 (c = 0.5, CH_3OH). IR (KBr): 3389, 3277, 3061, 2953, 2859, 1678, 1595, 1493, 1443, 1306, 1215, 1090, 841, 797, 745, 700 cm^{-1}. ESI-MS (*m/e*) 512 $[M-H]^-$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.79 (s, 1H), 10.68 (s, 1H), 10.42 (s, 1H), 8.34 (s, 1H), 7.55 (s, 1H), 7.44 (dd, *J* = 19.2 Hz, *J* = 7.2 Hz, 2H), 7.22 (d, *J* = 11.4 Hz, 1H), 7.06 (t, *J* = 6.3 Hz, 1H), 7.09–6.76 (m, *J* = 10.8 Hz, *J* = 7.2 Hz, 7H), 5.24 (s, 1H), 4.22 (d, *J* = 7.8 Hz, 1H), 3.83 (s, 3H), 3.59 (t, *J* = 6.9 Hz, 1H), 3.46 (dd, *J* = 17.1 Hz, *J* = 5.4 Hz, 1H), 3.31 (dd, *J* = 15.0Hz, *J* = 9.0 Hz, 1H), 2.82–2.80 (m, 2H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 174.08, 171.13, 169.65, 159.53, 136.49, 136.35, 134.44, 134.40, 129.70, 128.88, 128.49, 128.21, 127.26, 121.39, 118.81, 118.20, 117.59, 112.79, 111.54, 108.46, 59.15, 53.42, 52.88, 49.82, 46.46, 29.15, 24.77. Anal. Calcd for $C_{29}H_{27}N_3O_6$: C 67.83, H 5.30, N 8.18. Found: C 68.02, H 5.13, N 8.42. HPLC purity: 95%.

4.30. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-trp (4g)

Yield: 58 mg (34%). Mp 115–116 °C. $[\alpha]_D^{25}$ = −68.3 (c = 0.7, CH_3OH). IR (KBr): 3395, 3229, 3059, 2953, 2855, 1678, 1595, 1443, 1308, 1217, 1092, 839, 797, 745 cm^{-1}. ESI-MS (*m/e*) 551 $[M-H]^-$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.95 (s, 1H), 10.89 (s, 1H), 10.61 (s, 1H), 8.16 (s, 1H), 7.64 (s, 1H), 7.51 (t, *J* = 8.1 Hz, 2H), 7.36–7.29 (m, 3H), 7.17 (s, 1H), 7.10–6.91 (m, 6H), 5.47 (s, 1H), 4.51 (s, 1H), 3.82 (s, 3H), 3.17 (t, *J* = 5.4 Hz, 3H), 2.85 (dd, *J* = 15.3 Hz, *J* = 9.0 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.40, 171.13, 169.43, 160.15, 136.98, 136.74, 136.56, 127.71, 126.73, 124.21, 121.78, 121.40, 119.13, 118.85, 118.73, 118.30, 117.88, 113.23, 111.86, 109.87, 107.82, 53.68, 53.43, 52.91, 51.65, 27.46, 24.61. Anal. Calcd for $C_{31}H_{28}N_4O_6$: C 67.38, H 5.11, N 10.14. Found: C 67.15, H 5.26, N 10.37. HPLC purity: 93.3%.

4.31. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-tyr (4h)

Yield: 89 mg (52%). Mp 134–135 °C. $[\alpha]_D^{25}$ = −28.3 (c = 0.5, CH_3OH). IR (KBr): 3364, 3291, 3061, 2953, 2859, 1678, 1595, 1514, 1445, 1308, 1219, 1092, 837, 795, 745 cm^{-1}. ESI-MS (*m/e*) 528 $[M-H]^-$. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.68 (s, 1H), 8.20 (s, 1H), 8.01 (d, *J* = 7.2 Hz, 1H), 7.61 (s, 1H), 7.45 (d, *J* = 7.2 Hz, 1H), 7.38 (d, *J* = 8.4 Hz, 1H), 7.27 (d, *J* = 7.2 Hz, 1H), 7.08–6.99 (m, 5H), 6.68 (d, *J* = 7.2 Hz, 2H), 5.12 (s, 1H), 4.34 (s, 1H), 3.84 (s, 3H), 3.51 (s, 1H), 2.98–2.87 (m, 3H), 2.79 (dd, *J* = 15.0 Hz, *J* = 8.4 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.39, 172.76, 169.67, 159.75, 156.36, 136.58, 136.42, 134.27, 134.00, 130.63, 129.82, 127.86, 127.14, 127.10, 121.43, 118.87, 118.16, 117.71, 115.44, 112.84, 111.53, 108.27, 54.07, 53.40, 52.90, 52.43, 36.33, 24.53. Anal. Calcd for $C_{29}H_{27}N_3O_7$: C 65.78, H 5.14, N 7.94. Found: C 65.57, H 4.98, N 7.70. HPLC purity: 97.6%.

4.32. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-ser (4i)

Yield: 86 mg (52%). Mp 125–126 °C. $[\alpha]_D^{25} = -1.5$ (c = 0.55, CH_3OH: THF = 1 : 2). IR (KBr): 3383, 3298, 3063, 2955, 2853, 1678, 1595, 1493, 1447, 1310, 1215, 1090, 841, 797, 746 cm^{-1}. ESI-MS (*m/e*) 452 [M − H]⁻. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.73 (s, 1H), 8.17 (s, 1H), 8.08 (d, *J* = 7.5 Hz, 1H), 7.63 (s, 1H), 7.46 (t, *J* = 9.0 Hz, 2H), 7.28 (d, *J* = 8.1 Hz, 1H), 7.07 (t, *J* = 7.5 Hz, 1H), 6.99 (d, *J* = 8.4 Hz, 2H), 5.26 (s, 1H), 4.26 (s, 1H), 3.83 (s, 3H), 3.74 (dd, *J* = 10.8 Hz, *J* = 4.5 Hz, 1H), 3.64 (dd, *J* = 10.2 Hz, *J* = 2.7 Hz, 1H), 3.57 (dd, *J* = 8.1 Hz, *J* = 5.1 Hz, 1H), 3.03 (dd, *J* = 15.3 Hz, *J* = 4.2 Hz, 1H), 2.78 (dd, *J* = 15.0 Hz, *J* = 9.0 Hz, 1H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 172.77, 172.25, 169.65, 159.73, 136.50, 136.44, 134.22, 133.93, 129.93, 127.10, 121.48, 118.91, 118.20, 117.73, 112.88, 111.53, 108.48, 54.75, 53.47, 52.90, 51.98, 48.97, 24.85. Anal. Calcd for $C_{23}H_{23}N_3O_7$: C 60.92, H 5.11, N 9.27. Found: C 60.70, H 4.94, N 9.03. HPLC purity: 98.9%.

4.33. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-asn (4j)

Yield: 85 mg (51%). Mp 102–103 °C. $[\alpha]_D^{25} = -50.6$ (c = 0.65, CH_3OH). IR (KBr): 3362, 3204, 3063, 2955, 2857, 1676, 1597, 1493, 1445, 1310, 1217, 1092, 841, 795, 746 cm^{-1}. ESI-MS (*m/e*) 479 [M − H]⁻. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.81 (s, 1H), 10.60 (s, 1H), 8.47 (d, *J* = 9.6 Hz, 1H), 8.15 (s, 1H), 7.67 (s, 1H), 7.50–7.41 (m, 3H), 7.29 (d, *J* = 7.8 Hz, 1H), 7.09–6.95 (m, 4H), 5.43 (s, 1H), 4.53 (s, 1H), 3.84 (s, 3H), 3.16–3.10 (m, 2H), 2.88 (dd, *J* = 17.5 Hz, *J* = 7.8 Hz, 1H), 2.57 (s, 2H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 172.91, 171.71, 169.40, 163.46, 160.10, 136.89, 136.75, 136.59, 130.87, 126.69, 121.84, 119.15, 118.30, 117.84, 113.30, 111.71, 107.81, 53.55, 52.94, 51.72, 48.98, 36.88, 24.27. Anal. Calcd for $C_{24}H_{24}N_4O_7$: C 59.99, H 5.03, N 11.66. Found: C 59.77, H 5.19, N 11.41. HPLC purity: 94.9%.

4.34. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-gln (4k)

Yield: 96 mg (57%). Mp 116–117 °C. $[\alpha]_D^{25} = -40.7$ (c = 1.05, CH_3OH). IR (KBr): 3347, 3210, 3063, 2955, 2864, 1674, 1595, 1493, 1308, 1215, 1092, 841, 797, 745 cm^{-1}. ESI-MS (*m/e*) 493 [M − H]⁻. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.75 (s, 1H), 8.21 (s, 1H), 8.09 (d, *J* = 6.0 Hz, 1H), 7.62 (s, 1H), 7.46 (d, *J* = 7.2 Hz, 2H), 7.28 (d, *J* = 7.5 Hz, 1H), 7.05 (d, *J* = 7.8 Hz, 1H), 6.99 (d, *J* = 7.5 Hz, 2H), 6.79 (s, 2H), 5.23 (s, 1H), 4.15 (s, 1H), 3.83 (s, 3H), 3.49 (s, 1H), 3.01 (d, *J* = 12.3 Hz, 1H), 2.75 (dd, *J* = 15.0 Hz, *J* = 9.0 Hz, 1H), 2.12 (s, 2H), 1.99–1.83 (m, 2H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 174.01, 173.76, 172.93, 169.67, 159.67, 136.58, 136.46, 134.37, 134.22, 129.80, 127.10, 121.44, 118.90, 118.16, 117.71, 112.84, 111.53, 108.41, 53.44, 52.90, 52.12, 51.96, 31.74, 27.47, 25.19. Anal. Calcd for $C_{25}H_{26}N_4O_7$: C 60.72, H 5.30, N 11.33. Found: C 60.50, H 5.14, N 11.10. HPLC purity: 98.1%.

4.35. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-thr (4l)

Yield: 106 mg (63%). Mp 124–125 °C. $[\alpha]_D^{25} = -47.9$ (c = 0.85, CH_3OH). IR (KBr): 3368, 3283, 3063, 2976, 2878, 1678, 1595, 1493, 1443, 1308, 1215, 1090, 839, 795, 745 cm^{-1}. ESI-MS (*m/e*) 466 [M − H]⁻. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.73 (s, 1H), 8.20 (s, 1H), 7.89 (d, *J* = 8.1 Hz, 1H), 7.65 (s, 1H), 7.47 (t, *J* = 6.0 Hz, 2H), 7.27 (d, *J* = 7.8 Hz, 1H), 7.06 (t, *J* = 4.5 Hz, 1H), 6.99 (d, *J* = 8.1 Hz, 2H), 5.25 (s, 1H), 4.15 (s, 2H), 3.83 (s, 3H), 3.58 (t, *J* = 6.0 Hz, 1H), 3.04 (dd, *J* = 15.6 Hz, *J* = 4.5 Hz, 1H), 2.82 (dd, *J* = 15.3 Hz, *J* = 8.7 Hz, 1H), 1.06 (d, *J* = 6.0 Hz, 3H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 172.96, 172.54, 169.64, 159.74, 136.60, 136.44, 134.39, 133.94, 129.87, 127.14, 121.45, 118.88, 118.22, 117.78, 112.91, 111.51, 108.54, 65.39, 57.65, 53.58, 52.91, 52.40, 24.72, 20.90. Anal. Calcd for $C_{24}H_{25}N_3O_7$: C 61.66, H 5.39, N 8.99. Found: C 61.85, H 5.54, N 8.74. HPLC purity: 93.4%.

4.36. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-asp (4 m)

Yield: 68 mg (47%). Mp 115–116 °C. $[\alpha]_D^{25} = -40.5$ (c = 0.65, CH_3OH). IR (KBr): 3372, 3063, 2957, 2859, 1682, 1593, 1443, 1310, 1219, 1090, 839, 795, 745 cm^{-1}. ESI-MS (*m/e*) 480 [M − H]⁻. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.73 (s, 1H), 8.24 (d, *J* = 6.6 Hz, 1H), 8.18 (s, 1H), 7.63 (s, 1H), 7.46 (t, *J* = 9.0 Hz, 2H), 7.27 (d, *J* = 7.5 Hz, 1H), 7.07 (t, *J* = 7.5 Hz, 1H), 6.99 (d, *J* = 6.3 Hz, 2H), 5.26 (s, 1H), 4.38 (s, 1H), 3.83 (s, 3H), 3.04 (d, *J* = 12.3 Hz, 1H), 2.80 (dd, *J* = 15.0 Hz, *J* = 8.7 Hz, 1H), 2.60 (s, 2H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.05, 172.59, 172.06, 169.60, 159.82, 136.58, 136.47, 133.82, 133.35, 130.11, 127.01, 121.54, 118.95, 118.22, 117.76, 112.98, 111.57, 108.30, 53.50, 52.90, 52.05, 48.84, 37.96, 24.70. Anal. Calcd for $C_{24}H_{23}N_3O_8$: C 59.87, H 4.82, N 8.73. Found: C 59.65, H 4.67, N 8.51. HPLC purity: 95.4%.

4.37. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-glu (4n)

Yield: 82 mg (57%). Mp 118–119 °C. $[\alpha]_D^{25} = -13.1$ (c = 0.45, CH_3OH). IR (KBr): 3364, 3213, 3065, 2955, 2874, 1678, 1593, 1445, 1308, 1217, 1090, 837, 795, 746 cm^{-1}. ESI-MS (*m/e*) 494 [M − H]⁻. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.75 (s, 1H), 8.24 (s, 1H), 8.07 (d, *J* = 7.2 Hz, 1H), 7.62 (s, 1H), 7.45 (s, 2H), 7.27 (d, *J* = 7.5 Hz, 1H), 7.06 (t, *J* = 6.9 Hz, 1H), 6.98 (d, *J* = 8.1 Hz, 2H), 5.21 (s, 1H), 4.17 (s, 1H), 3.83 (s, 3H), 3.49 (s, 1H), 3.00 (dd, *J* = 15.0 Hz, *J* = 3.3 Hz, 1H), 2.75 (dd, *J* = 15.0 Hz, *J* = 8.7 Hz, 1H), 2.27 (s, 2H), 1.90 (m, *J* = 7.2 Hz, 2H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 174.45, 173.69, 172.77, 169.65, 159.68, 136.41, 134.40, 134.10, 129.81, 127.11, 121.42, 118.88, 118.17, 117.71, 112.86, 111.52, 108.40, 53.47, 52.88, 52.05, 51.90, 30.48, 27.29, 25.09. Anal. Calcd for $C_{25}H_{25}N_3O_8$: C 60.60, H 5.09, N 8.48. Found: C 60.82, H 5.24, N 8.25. HPLC purity: 96.5%.

4.38. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-lys (4°)

Yield: 85 mg (62%). Mp 117–118 °C. $[\alpha]_D^{25} = -50.1$ (c = 0.8, CH_3OH). IR (KBr): 3364, 3201, 3063, 2951, 2868, 1678, 1587, 1493, 1445, 1383, 1319, 1215, 1092, 839, 795, 746 cm^{-1}. ESI-MS (*m/e*) 493 [M − H]⁻. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.77 (s, 1H), 8.30 (s, 3H), 7.98 (d, *J* = 4.5 Hz, 1H), 7.60 (s, 1H), 7.45 (t, *J* = 9.0 Hz, 2H), 7.28 (d, *J* = 7.2 Hz, 1H), 7.06–7.00 (m, 3H), 5.21 (s, 1H), 4.05 (s, 1H), 3.82 (s, 3H), 3.46 (s, 1H), 3.02 (d, *J* = 13.2 Hz, 1H), 2.72 (s, 3H), 1.72–1.09 (m, 6H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 174.32, 172.47, 169.60, 159.63, 136.43, 136.32, 134.50, 134.23, 129.77, 127.14, 121.42, 118.86, 118.17, 117.75, 112.92, 111.52, 108.59, 53.57, 52.88, 52.13, 38.85, 31.64, 27.13, 25.25, 22.54. Anal. Calcd for $C_{26}H_{30}N_4O_6$: C 63.15, H 6.11, N 11.33. Found: C 63.34, H 6.27, N 11.56. HPLC purity: 94.4%.

4.39. N-[(1R,3S)-1-(4-hydroxy-3-methoxycabonylphenyl)-2,3,4,9-tetrahydro-β-carboli-ne-3-formyl]-L-arg (4p)

Yield: 81 mg (47%). Mp 122–124 °C. $[\alpha]_D^{25} = -49.5$ (c = 0.9, CH_3OH). IR (KBr): 3333, 3229, 3065, 2953, 2878, 1678, 1595, 1493, 1445, 1269, 1217, 1092, 839, 746, 694 cm^{-1}. ESI-MS (*m/e*) 521 [M − H]⁻. ^1H NMR (300 MHz, DMSO-d_6) δ/ppm = 10.75 (s, 1H), 8.17 (s, 3H), 7.62 (s, 1H), 7.46 (t, *J* = 9.9 Hz, 2H), 7.28 (d, *J* = 8.1 Hz, 1H), 7.07 (t, *J* = 7.5 Hz, 1H), 7.00 (d, *J* = 8.7 Hz, 2H), 5.27 (s, 1H), 4.19 (s, 1H), 3.85 (s, 3H), 3.58 (dd, *J* = 7.8 Hz, *J* = 4.8 Hz, 1H), 3.15 (s, 2H), 3.03 (dd, *J* = 15.3 Hz, *J* = 4.5 Hz, 1H), 2.78 (dd, *J* = 15.0 Hz, *J* = 9.0 Hz, 1H), 1.76 (t, *J* = 6.3 Hz, 1H), 1.65 (t, *J* = 7.2 Hz, 1H), 1.52 (s, 2H); ^{13}C NMR (75 MHz, DMSO-d_6) δ/ppm = 173.79, 172.77, 169.60, 163.71, 159.74, 136.61, 136.51, 136.45, 130.01, 127.01, 121.52, 118.95, 118.19, 117.74,

112.94, 111.56, 108.22, 65.38, 53.47, 52.90, 52.03, 51.94, 28.88, 25.10, 15.62. Anal. Calcd for $C_{26}H_{30}N_6O_6$: C 59.76, H 5.79, N 16.08. Found: C 59.97, H 5.94, N 6.32. HPLC purity: 92.6%.

4.40. Animals

Male SD (Sprague Dawley) rats weighing 250–300 g, purchased from Peking University Health Science Center, were maintained at 21 °C with a natural day/night cycle in a conventional animal colony. The assessments described here were performed based on a protocol reviewed and approved by the ethics committee of Capital Medical University. The committee assures the animals' welfare to be was maintained in accordance with the requirements of the animal welfare act and according to the guide for care and use of laboratory animals.

4.41. In vitro anti-platelet aggregation activity assay

An H-10 cell counter was used to determine the platelet count and a two channel chronology aggregometer was used to evaluate platelet aggregation. The pig blood (six pigs, purchased from Animal Center of Peking University) was centrifuged at 950 g for 10 min to obtain the platelet rich plasma (PRP). The remaining blood was centrifuged for an additional 10 min at 1500 g to get platelet poor plasma (PPP). The final platelet count of the PRP was adjusted to 2×10^8 platelets/ml with autologous PPP. 0.5 ml of the adjusted plasma sample and 5 μl of NS or 5 μl of the solution of **4a-p** [the final concentrations were 500, 100, 10, 1, 0.1, 0.01 and 0.001 μM, which were prepared by diluting the stock solutions of 10 mM of **4a-p** in DMSO/NS, 1/10 (v/v) with NS] was added to an optical aggregometry testing tube. After the adjustment of the baseline, 5 μl of the solution was added in which may be optional by 5 μl of PAF in NS (final concentration 0.1 μM) or 5 μl of the solution of ADP in NS (final concentration 10 μM) or 5 μl of the solution of AA in 1 M aqueous sodium carbonate (final concentration 350 μM) or 50 μl of the solution of TH in NS (final concentration 0.1U/ml). And the aggregation was measured at 37 °C for 5 min. The effects of **4a-p** (at a series of concentrations ranging from 500 μM to 1 nM) on the platelet aggregation induced by PAF, ADP, AA or TH could be observed. All these antiplatelet aggregation tests in six duplicate tubes were carried out. The maximum platelet aggregation (Am) of control group (NS) or **4a-p** group was represented by the peak height of aggregation curve (equal to the maximum light transmission). The inhibition rate was calculated according to the formula: inhibition (%) = [(Am of NS) - (Am of **4a-p**)]/(Am of NS) × 100%. Am of NS stands for the values of platelet aggregation induced by PAF, ADP, AA, and TH without **4a-p** and the values were 56.52 ± 4.59, 52.07 ± 3.81, 50.94 ± 3.75, and 59.66 ± 4.95, respectively. The concentration vs inhibition rate curve was plotted to determine the IC_{50} value with GWBASIC.EXE program.

4.42. In vivo anti-thrombotic assay of oral administration of 4a-p in rat model

The assessments described here were performed based on a protocol reviewed and approved by the ethics committee of Capital Medical University. The committee assures that the welfare of the animals was maintained in accordance with the requirements of the animal welfare act and with the guide for care and use of laboratory animals. Aspirin and **4a-p** were suspended or dissolved in NS before oral administration and kept in an ice bath. Male Wister rats weighing 250–300 g (purchased from Animal Center of Peking University) were used. The rats were fed with the suspension of aspirin in NS (dosage 167 μmol/kg and 16.7) or **4a-p** in NS (dosage, 1 nmol/kg) or NS alone, then anesthetized with pentobarbital sodium (80.0 mg/kg, i.p.) before separating its the right carotid artery and left jugular vein. A weighed 6-cm long thread was inserted into the middle of a polyethylene tube. The polyethylene tube was filled with heparin sodium (50 IU/ml in NS) with its one end being inserted into the left jugular vein. Heparin sodium was injected as anticoagulant from the other end of the polyethylene tube, which was inserted into the right carotid artery. Blood was allowed to flow from the right carotid artery to the left jugular vein through the polyethylene tube for 15 min. The thread was removed to obtain the weight of the wet thrombus.

4.43. QSAR module of Cerius2 based MFA of 4a-p

MFA (implemented in QSAR module of Cerius2) studies were performed for 4a-p based on the conformational alignment using alignment methods common substructure (CSS) for MFA studies. MFA is a method for quantifying the interaction energy between a probe molecule and a set of aligned target molecules [51–53]. A five-step procedure consisted of building molecules, energy minimization, forming molecular fields, matching atoms and aligning molecules, selecting probes, and generating the QSAR equations was practiced in MFA. (1*R*,3*S*)-THCCA was selected as a standard reference (target model) with which the conformations of 4a-p were aligned through superpositioning. The Align method - RMS atoms and Align type - Flexible were used. A rectangular field was generated using the probes proton (H+), hydroxyl group (OH-) and methyl group (CH_3). A grid spacing of 2.5 Å was used, and fields at 637 points were generated. The energy cutoff was kept at −20 to +20 Kcal. Only the top 10% of the total descriptors, which explained the maximum variance, were used in the generation of the QSAR. Regression analysis was carried out using the genetic partial least-squares (G/PLS) method consisting of 50000 generations with a population size of 100. The optimal number of components was set to 5.

4.44. Dose-effect relationship of 4j for thrombogenesis in rat model

Aspirin, **2** and **4j** were suspended in NS before oral administration and kept in an ice bath. Male Wister rats weighing 250–300 g (purchased from Animal Center of Peking University) were used. The rats were fed with the suspension of aspirin in NS (dosage, 165 μmol/kg) or **4j** in NS (dosage, 1, 0.01 and 0.001 nmol/kg) or **2** in NS (dosage, 10 nmol/kg) or NS alone, then anesthetized with pentobarbital sodium (80.0 mg/kg, i.p.) before the separation of its right carotid artery and left jugular vein. A weighed 6-cm long thread was inserted into the middle of a polyethylene tube. The polyethylene tube was filled with heparin sodium (50 IU/ml in NS) with its one end being inserted into the left jugular vein. Heparin sodium was injected as anticoagulant from the other end of the polyethylene tube which was inserted into the right carotid artery. Blood was allowed to flow from the right carotid artery to the left jugular vein through the polyethylene tube for 15 min. The thread was removed to obtain the weight of the wet thrombus.

4.45. Cell culture

L02, NCM460 and NHDF cells (purchased from KeyGen Biotech Co., Ltd. and Bena Culture Collection) were cultured in RPMI-1640 or DMEM medium (KeyGen Biotech Co.,Ltd., KGM31800-500, KGM12800-500). The medium was supplemented with 10% fetal bovine serum (FBS) (Hyclone, SH30084.03). Cells were cultured in an incubator containing 5% CO_2 at 37 °C.

4.46. Cell proliferation assay

Anti-proliferative activity was measured using the Cell-Counting Kit 8 (PBM, 230801) according to the manufacturer's protocol. RPMI 1640 or DMEM containing 10% FBS and 1% penicillin/streptomycin was used as the cell viability assay medium. To test the anti-proliferative activity of 4j, cells were seeded at 5000 cells per well in 96-well plate. Six hours after seeding, cells were treated with 4j (two-fold dilution, six concentrations). After 48 h of treatment, cell were incubated with a 10% CCK-8 solution at 37 °C for 1 h, and IC_{50} values were calculated using the percentage of growth of treated cells versus the DMSO control. The

results were analyzed using GraphPad Prism 7.0.

4.47. Rat tail bleeding time assay

Rat was anesthetized with isoflurane anesthesia (~2.5% vol/vol) mixed with air (~0.4 L/min) and then subjected to surgical dissection of the tail (3 mm from the tip). The tail was prewarmed in 37 °C NS for 2 min before dissection. The time of bleeding was recorded. Experiments were performed in thrombosis model of rats orally treating with NS (3 mL/kg), aspirin in NS (167 μmol/kg), or 4j in NS (1 nmol/kg, 0.01 nmol/kg and 0.001 nmol/kg).12 rats were included in each group.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors of this paper extent their appreciation to the Special Project of China (2018ZX097201003) and the National Natural Science Foundation of China, 81703332 for their financial support.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ejmech.2022.114681.

References

[1] Y. Matsuo, T. Mizuochi, M. Miho, S. Nakagawa, S. Ozono, K. Ueda, Y. Sogabe, R. Seki, K. Soejima, T. Okamura, Y. Yamashita, Factor X deficiency with heterozygous mutations of novel p.G435S and known p.G244R in a patient presenting with severe umbilical hemorrhage, Kurume Med. J. 63 (2017) 23–28.
[2] W.E. Winter, D.N. Greene, S.G. Beal, J.A. Isom, H. Manning, G. Wilkerson, N. Harris, Clotting factors: clinical biochemistry and their roles as plasma enzymes, Adv. Clin. Chem. 94 (2020) 31–84.
[3] K.C. Childers, S.C. Peters, P. Lollar, H.T. Spencer, C.B. Doering, P.C. Spiegel, SAXS Analysis of the Intrinsic Tenase Complex Bound to a Lipid Nanodisc Highlights Intermolecular Contacts between Factors VIIIa/IXa, Blood advances, 2022.
[4] O.M. Al-Amer, The Role of Thrombin in Haemostasis, Blood Coagulation & Fibrinolysis : an International Journal in Haemostasis and Thrombosis, 2022.
[5] R. Lordan, A. Tsoupras, I. Zabetakis, Investigation of platelet aggregation in atherosclerosis, methods in molecular biology, Clifton, N.J.) 2419 (2022) 333–347.
[6] E. Paszek, E. Pociask, M. Ząbczyk, A. Piórkowski, S. Butenas, J. Legutko, A. Undas, Active factor XI is associated with the risk of cardiovascular events in stable coronary artery disease patients, Atherosclerosis 346 (2022) 124–132.
[7] R. Choxi, K. Kapoor, N. Mackman, I.S. Jovin, Direct Oral Anticoagulants and Coronary Artery Disease, Arteriosclerosis, Thrombosis, and Vascular Biology, 2022. Atvbaha121317171.
[8] R.A. Al-Horani, A.Y. Mehta, U.R. Desai, Potent direct inhibitors of factor Xa based on the tetrahydroisoquinoline scaffold, Eur. J. Med. Chem. 54 (2012) 771–783.
[9] M. Ilić, D. Kikelj, J. Ilaš, Fluorinated dual antithrombotic compounds based on 1,4-benzoxazine scaffold, Eur. J. Med. Chem. 50 (2012) 255–263.
[10] D. Chen, J. Shi, J. Liu, X. Zhang, X. Deng, Y. Yang, S. Cui, Q. Zhu, G. Gong, Y. Xu, Design, synthesis and antithrombotic evaluation of novel non-peptide thrombin inhibitors, Bioorg. Med. Chem. 25 (2017) 458–470.
[11] H. Yoo, S.K. Ku, W. Lee, S. Kwak, Y.D. Baek, B.W. Min, G.S. Jeong, J.S. Bae, Antiplatelet, anticoagulant, and profibrinolytic activities of cudratricusxanthone A, Arch Pharm. Res. (Seoul) 37 (2014) 1069–1078.
[12] K. Takeuchi, J.A. Bastian, D.S. Gifford-Moore, R.W. Harper, S.C. Miller, J. T. Mullaney, D.J. Sall, G.F. Smith, M. Zhang, M.J. Fisher, 1,2-Disubstituted indole, azaindole and benzimidazole derivatives possessing amine moiety: a novel series of thrombin inhibitors, Bioorg. Med. Chem. Lett 10 (2000) 2347–2351.
[13] P.E. Sanderson, M.G. Stanton, B.D. Dorsey, T.A. Lyle, C. McDonough, W. M. Sanders, K.L. Savage, A.M. Naylor-Olsen, J.A. Krueger, S.D. Lewis, B.J. Lucas, J. J. Lynch, Y. Yan, Azaindoles: moderately basic P1 groups for enhancing the selectivity of thrombin inhibitors, Bioorg. Med. Chem. Lett 13 (2003) 795–798.
[14] J.Z. Deng, D.R. McMasters, P.M. Rabbat, P.D. Williams, C.A. Coburn, Y. Yan, L. C. Kuo, S.D. Lewis, B.J. Lucas, J.A. Krueger, B. Strulovici, J.P. Vacca, T.A. Lyle, C. S. Burgey, Development of an oxazolopyridine series of dual thrombin/factor Xa inhibitors via structure-guided lead optimization, Bioorg. Med. Chem. Lett 15 (2005) 4411–4416.
[15] E.L. Wu, Y. Mei, K. Han, J.Z. Zhang, Quantum and molecular dynamics study for binding of macrocyclic inhibitors to human alpha-thrombin, Biophys. J. 92 (2007) 4244–4253.
[16] T. Lu, T. Markotan, S.K. Ballentine, E.C. Giardino, J. Spurlino, C.S. Crysler, K. Brown, B.E. Maryanoff, B.E. Tomczuk, B.P. Damiano, U. Shukla, D. End, P. Andrade-Gordon, R.F. Bone, M.R. Player, Discovery and clinical evaluation of 1-{N-[2-(amidinoaminooxy)ethyl]amino}carbonylmethyl-6-methyl-3-[2,2-difluoro-2-phenylethylamino]pyrazinone (RWJ-671818), a thrombin inhibitor with an oxyguanidine P1 motif, J. Med. Chem. 53 (2010) 1843–1856.
[17] R. Purgatorio, N. Gambacorta, M. de Candia, M. Catto, M. Rullo, L. Pisani, O. Nicolotti, C.D. Altomare, First-in-Class isonipecotamide-based thrombin and cholinesterase dual inhibitors with potential for alzheimer disease, Molecules (2021) 26. Basel, Switzerland).
[18] Y. Lei, B. Zhang, Y. Zhang, X. Dai, Y. Duan, Q. Mao, J. Gao, Y. Yang, Z. Bao, X. Fu, K. Ping, C. Yan, Y. Mou, S. Wang, Design, synthesis and biological evaluation of novel FXIa inhibitors with 2-phenyl-1H-imidazole-5-carboxamide moiety as P1 fragment, Eur. J. Med. Chem. 220 (2021), 113437.
[19] K. Takeuchi, T.J. Kohn, R.W. Harper, H.S. Lin, D.S. Gifford-Moore, M.E. Richett, D. J. Sall, G.F. Smith, M. Zhang, Diamino benzo[b]thiophene derivatives as a novel class of active site directed thrombin inhibitors. Part 6: further focus on the contracted C4'-side chain analogues, Bioorg. Med. Chem. Lett 10 (2000) 1199–1202.
[20] B. Baum, L. Muley, A. Heine, M. Smolinski, D. Hangauer, G. Klebe, Think twice: understanding the high potency of bis(phenyl)methane inhibitors of thrombin, J. Mol. Biol. 391 (2009) 552–564.
[21] T. Ohmura, I. Takahashi, K. Ikegami, J. Encinas, [Pharmacological and clinical profiles of the direct thrombin inhibitor dabigatran etexilate methane sulfate (Prazaxa(®))], Nihon yakurigaku zasshi, Folia Pharmacol. Jpn. 138 (2011) 79–88.
[22] C.C. Yang, L.D. Hsiao, H.H. Lin, H.C. Tseng, J.H. Situmorang, Y.L. Leu, C.M. Yang, Induction of HO-1 by 5, 8-dihydroxy-4',7-dimethoxyflavone via activation of ROS/p38 MAPK/Nrf2 attenuates thrombin-induced connective tissue growth factor expression in human cardiac fibroblasts, Oxid. Med. Cell. Longev. 2020 (2020), 1080168.
[23] S. Chen, K. Lv, A. Sharda, J. Deng, W. Zeng, C. Zhang, Q. Hu, P. Jin, G. Yao, X. Xu, Z. Ming, C. Fang, Anti-thrombotic effects mediated by dihydromyricetin involve both platelet inhibition and endothelial protection, Pharmacol. Res. 167 (2021), 105540.
[24] M. Hrubša, R. Alva, M.S. Parvin, K. Macáková, J. Karlíčková, J. Fadraersada, L. Konečný, M. Moravcová, A. Carazo, P. Mladěnka, Comparison of antiplatelet effects of phenol derivatives in humans, Biomolecules 12 (2022).
[25] A. Hillisch, K.M. Gericke, S. Allerheiligen, S. Roehrig, M. Schaefer, A. Tersteegen, S. Schulz, P. Lienau, M. Gnoth, V. Puetter, R.C. Hillig, S. Heitmeier, Design, synthesis, and pharmacological characterization of a neutral, non-prodrug thrombin inhibitor with good oral pharmacokinetics, J. Med. Chem. 63 (2020) 12574–12594.
[26] B.D. Belviso, R. Caliandro, M. de Candia, G. Zaetta, G. Lopopolo, F. Incampo, M. Colucci, C.D. Altomare, How a β-D-glucoside side chain enhances binding affinity to thrombin of inhibitors bearing 2-chlorothiophene as P1 moiety: crystallography, fragment deconstruction study, and evaluation of antithrombotic properties, J. Med. Chem. 57 (2014) 8563–8575.
[27] N.H. Hauel, H. Nar, H. Priepke, U. Ries, J.M. Stassen, W. Wienen, Structure-based design of novel potent nonpeptide thrombin inhibitors, J. Med. Chem. 45 (2002) 1757–1766.
[28] X. Liao, X. Zhuang, C. Liang, J. Li, R. Flaumenhaft, C. Yuan, M. Huang, Flavonoids as protein disulfide isomerase inhibitors: key molecular and structural features for the interaction, J. Agric. Food Chem. 70 (14) (2022) 4475–4483.
[29] K.Z. Du, Y. Cui, S. Chen, R. Yang, Y. Shang, C. Wang, Y. Yan, J. Li, Y.X. Chang, An integration strategy combined progressive multivariate statistics with anticoagulant activity evaluation for screening anticoagulant quality markers in Chinese patent medicine, J. Ethnopharmacol. 287 (2022), 114964.
[30] Z. Tang, Y. Ren, F. Liu, Identify thrombin inhibitor with novel skeleton based on virtual screening study, J. Biomol. Struct. Dyn. 40 (2022) 499–507.
[31] A.P. Schaffner, P. Sansilvestri-Morel, N. Despaux, E. Ruano, T. Persigand, A. Rupin, P. Mennecier, M.O. Vallez, E. Raimbaud, P. Desos, P. Gloanec, Phosphinanes and azaphosphinanes as potent and selective inhibitors of activated thrombin-activatable fibrinolysis inhibitor (TAFIa), J. Med. Chem. 64 (2021) 3897–3910.
[32] J. Kolodziejczyk-Czepas, M. Ponczek, M. Sady-Janczak, R. Pilarski, B. Bukowska, Extracts from Uncaria tomentosa as antiplatelet agents and thrombin inhibitors - the in vitro and in silico study, J. Ethnopharmacol. 267 (2021), 113494.
[33] F. Chen, G. Huang, Mechanism and inhibition kinetics of peptide P13 as thrombin inhibitor, Int. J. Biol. Macromol. 150 (2020) 1046–1052.
[34] S. Zhao, Z. Li, F. Huang, J. Wu, L. Gui, X. Zhang, Y. Wang, X. Wang, S. Peng, M. Zhao, Nano-scaled MTCA-KKV: for targeting thrombus, releasing pharmacophores, inhibiting thrombosis and dissolving blood clots in vivo, Int. J. Nanomed. 14 (2019) 4817–4831.
[35] X. Zhang, Y. Zhang, Y. Wang, J. Wu, H. Chen, M. Zhao, S. Peng, Modifying ICCA with trp-phe-phe to enhance in vivo activity and form nano-medicine, Int. J. Nanomed. 15 (2020) 465–481.
[36] J. Wu, M. Zhao, Y. Wang, Y. Wang, H. Zhu, S. Zhao, S. Peng, A novel lead of P-selectin inhibitor: discovery, synthesis, bioassays and action mechanism, Bioorg. Med. Chem. Lett 26 (2016) 4631–4636.
[37] J. Wu, M. Zhao, Y. Wang, Y. Wang, H. Zhu, S. Zhao, L. Gui, X. Zhang, S. Peng, N-(3-hydroxymethyl-β-carboline-1-yl-ethyl- 2-yl)-l-Phe: development toward a

nanoscaled antitumor drug capable of treating complicated thrombosis and inflammation, Drug Des. Dev. Ther. 11 (2017) 225–239.
[38] J. Wu, Y. Cui, X. Zhang, L. Gui, Y. Wang, S. Peng, M. Zhao, BCESA: a nano-scaled intercalator capable of targeting tumor tissue and releasing anti-tumoral β-carboline-3-carboxylic acid, Int. J. Nanomed. 14 (2019) 3027–3041.
[39] Z. Li, F. Huang, J. Wu, L. Gui, X. Zhang, Y. Wang, S. Zhao, X. Wang, S. Peng, M. Zhao, Heptapeptide-based modification leading to enhancing the action of MTCA on activated platelets, P-selectin, GPIIb/IIIa, Future Med. Chem. 10 (2018) 1957–1970.
[40] M. Hou, M. Lv, X. Zhang, Y. Wang, S. Zhao, J. Wu, S. Peng, M. Zhao, Discovery of novel (6S/12aS)-heptachpyridone capable of inhibiting thrombosis in vivo, Bioorg. Med. Chem. Lett 30 (2020), 127440.
[41] H. Chen, A. Lu, X. Zhang, L. Gui, Y. Wang, J. Wu, H. Feng, S. Peng, M. Zhao, Design and development of ICCA as a dual inhibitor of GPIIb/IIIa and P-selectin receptors, Drug Des. Dev. Ther. 12 (2018) 2097–2110.
[42] M. Zhao, C. Wang, Y. Wu, K. Zhou, S. Peng, Synthesis and thrombolytic activity of carboline-3-carboxylic acid modified metabolites of Ala-Arg-Pro-Ala-Lys, Prep. Biochem. Biotechnol. 34 (2004) 57–76.
[43] M. Zhao, L. Bi, W. Bi, C. Wang, Z. Yang, J. Ju, S. Peng, Synthesis of new class dipeptide analogues with improved permeability and antithrombotic activity, Bioorg. Med. Chem. 14 (2006) 4761–4774.
[44] K. Yao, M. Zhao, X. Zhang, Y. Wang, L. Li, M. Zheng, S. Peng, A class of oral N-[(1S,3S)-1-methyl-1,2,3,4-tetrahydro-β-carboline-3-carbonyl]- N'-(amino-acid-acyl)hydrazine: discovery, synthesis, in vitro anti-platelet aggregation/in vivo anti-thrombotic evaluation and 3D QSAR analysis, Eur. J. Med. Chem. 46 (2011) 3237–3249.
[45] Y. Wu, M. Zhao, C. Wang, S. Peng, Synthesis and thrombolytic activity of pseudopeptides related to fibrinogen fragment, Bioorg. Med. Chem. Lett 12 (2002) 2331–2333.
[46] J. Liu, G. Wu, G. Cui, W.X. Wang, M. Zhao, C. Wang, Z. Zhang, S. Peng, A new class of anti-thrombosis hexahydropyrazino-[1',2':1,6]pyrido-[3,4-b]-indole-1,4-dions: design, synthesis, logK determination, and QSAR analysis, Bioorg. Med. Chem. 15 (2007) 5672–5693.
[47] J. Liu, X. Jiang, M. Zhao, X. Zhang, M. Zheng, L. Peng, S. Peng, A class of 3S-2-aminoacyltetrahydro-beta-carboline-3-carboxylic acids: their facile synthesis, inhibition for platelet activation, and high in vivo anti-thrombotic potency, J. Med. Chem. 53 (2010) 3106–3116.
[48] N. Lin, M. Zhao, C. Wang, S. Peng, Synthesis and antithrombotic activity of carbolinecarboxyl RGD sequence, Bioorg. Med. Chem. Lett 12 (2002) 585–587.
[49] L. Li, G. Cui, M. Zhao, Y. Wang, H. Wang, W. Li, S. Peng, Assembly of beta-cyclodextrin with 3S-tetrahydro-beta-carboline-3-carboxylic acid and self-assembly of 6-(3'S-carboline-3'-carboxylaminoethylamino)-6-deoxy-beta-cyclodextrin: approaches to enhance anti-oxidation stability and anti-thrombotic potency, J. Phys. Chem. B 112 (2008) 12139–12147.
[50] C. Li, X. Zhang, M. Zhao, Y. Wang, J. Wu, J. Liu, M. Zheng, S. Peng, A class of novel N-(1-methyl-β-carboline-3-carbonyl)-N'-(aminoacid-acyl)-hydrazines: aromatization leaded design, synthesis, in vitro anti-platelet aggregation/in vivo anti-thrombotic evaluation and 3D QSAR analysis, Eur. J. Med. Chem. 46 (2011) 5598–5608.
[51] P. Sharma, N. Ghoshal, Exploration of a binding mode of benzothiazol-2-yl acetonitrile pyrimidine core based derivatives as potent c-jun N-terminal kinase-3 inhibitors and 3D-QSAR analyses, J. Chem. Inf. Model. 46 (2006) 1763–1774.
[52] N. Ghoshal, P. Mukherjee, 3-D-QSAR of N-substituted 4-amino-3, 3-dialkyl-2(3H)-furanone GABA receptor modulators using molecular field analysis and receptor surface modelling study, Bioorg. Med. Chem. Lett 14 (2004) 103–109.
[53] P. Bhattacharya, J. T. Leonard, K. Roy, Exploring 3D-QSAR of thiazole and thiadiazole derivatives as potent and selective human adenosine A3 receptor antagonists, J. Mol. Model. (11) 2005 516-524.